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SOFTWARE REQUIREMENT SPECIFICATION FOR POPULATION COUNT SYSTEM

Version 1

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Submitted to :- Sinodos .G

Submitted Date : 25 /1/2026

TABLE OF CONTENTS

CHAPTER ONE: INTRODUCTION …… 5

1.1 Background of the Study ……5

1.2 Problem Statement …… 6

1.3 Objectives of the Project ……7

1.3.1 General Objective …….. 7

1.3.2 Specific Objectives …… 7

1.4 Scope of the Project …… 9

1.5 Significance of the Study …… 9

1.6 Project Methodology Overview …… 10

1.7 Organization of the Document …….. 12

1.8 Budget and Development Plan …… 12

CHAPTER TWO: EXISTING SYSTEM ANALYSIS …….. 14

2.1 Existing System Analysis …… 14

2.1.1 Description of the Existing System …… 14

2.1.2 Limitations of the Existing System …… 15

2.2 Requirements Engineering Process Models …… 17

2.3 Requirements Elicitation Techniques …… 19

2.4 Requirements Analysis and Negotiation …… 22

2.5 Requirements Specification Standards …… 24

2.6 Requirements Validation and Verification …… 25

2.7 Challenges in Software Requirements Engineering …… 27

2.8 Review of Related Systems …… 28

CHAPTER THREE: SYSTEM ANALYSIS 31
3.1 Overview of Software Requirements Engineering 31
3.2 Proposed System Overview 33
3.3 Stakeholder Identification and Analysis..... 34
3.4 Business Process Analysis 35
3.5 Feasibility Study 37
3.5.1 Technical Feasibility 37
3.5.2 Economic Feasibility 38
3.5.3 Operational Feasibility 38
3.6 Requirements Elicitation Techniques Used 39
3.7 System Requirements 40
3.7.1 Functional Requirement 41
3.7.2 Non-Functional Requirement 49
3.8 Assumptions and Constraints 53
3.9 Requirements Prioritization..... 54
3.10 Requirements Traceability 57

CHAPTER FOUR : SYSTEM MODELING AND DESIGN..... 61
4.1 System Architecture Overview 61
4.2 Use Case Diagram and Descriptions..... 63
4.3 Activity Diagrams 64
4.4 Sequence Diagrams 72
4.5 Class Diagram..... 85
4.6 Data Flow Diagram (DFD) 90
4.7 Entity Relationship Diagram (ERD) 93
4.8 User Interface Requirements 97
4.9 Logical Database Requirements 101
4.10 Quality Attributes 103
4.11 Requirements Traceability Matrix (RTM) 106

CHAPTER FIVE: CONCLUSION AND RECOMMENDATIONS ……..……..……108

5.1 Conclusion …….. 108

5.2 Recommendations ……. 109

Reference …….. 111

CHAPTER ONE

INTRODUCTION

1.1 Back-ground Study

- ✓ Population data is really important for a country to develop. We need to know how many people live in a place to make decisions. In Ethiopia it is very important to have the numbers of people for things, like schools, hospitals, houses, roads and food. The. People who make decisions use this information to create plans and see if they are working. Population data is necessary for Ethiopia to make sure everyone gets what they need.

- ✓ Population counting in Ethiopia is done mostly through censuses and manual records. These things happen every now and then. They take a lot of time. Need many people, a lot of money and equipment to get everything done. The problem is that the information about the population gets old fast. This is because the population is growing quickly people are moving to new places cities are getting bigger and some people have to leave their homes because of economic or environmental problems. Population counting in Ethiopia is still an issue. Population data is very important.. Population counting, in Ethiopia is not easy.

- ✓ The use of technologies in Ethiopia is getting bigger with more internet and computer systems being put in place and the government is using these technologies to do its work. This is a chance for Ethiopia to make the way it keeps track of its population better. If we use a computer program to count the population we can make sure the information is always up to date it will be more accurate. The government can get the information it needs when it needs it no matter what level of government it is from the local to the national level and this will be very helpful for population data management in Ethiopia and for the population count system, in Ethiopia.

- ✓ This project focuses on the development of a Population Count System that aligns with Ethiopia's current development context. The system aims to provide a centralized, secure, and scalable platform for managing population data, reducing dependency on manual processes, and supporting evidence-based planning and decision-making.

1.2 Problem Statement

- ✓ Population data that is accurate and current is really important for a country to plan properly and provide services to its people. In Ethiopia they usually get population information from national surveys that happen every now and then and from various government offices that keep their own records. The problem is that these methods are mostly done by hand they are not updated often and they need a lot of money and people to make them work. Population data is something that Ethiopia needs to get right so they can make decisions about services and planning and this is why good population data is so important, for the country.

- ✓ The big issue is that we do not have population data that's up to date. Because the population is growing fast and people are moving around a lot the information we have about how many people live in a place becomes old very quickly. This means that the numbers we use are not correct and that makes it hard to plan things like schools and hospitals and houses and roads. Population data is a problem because it is not real-time. We need population data to make sure that we can plan for things, like education and healthcare and housing and infrastructure development.

- ✓ The way we manage population data now is not very good. We have a lot of problems with population data not being correct and having duplicate information. Population records are usually kept in systems by the federal government by regions and by local areas. This makes it hard to combine all the data and make sure it is correct. Because we do not have one digital system population data is at risk of being incorrect because of human mistakes getting lost or people accessing it without permission.
- ✓ Population data is very important. We need to be careful, with it. Population data management needs to be improved so we can keep population data safe and accurate.

- ✓ Population data is really hard to get to and use. The people who make decisions and plans often have to wait a time to get good information, about the population. This is because the information is processed by hand and the different groups do not work together.
- ✓ As a result the people who make decisions do not have the information they need when they need it. This makes it harder to make good development policies. Population data is very important. It is a problem that population data is not easy to get to and use.

- ✓ Therefore, there is a critical need for a software-based Population Count System that can provide a centralized, secure, and continuously updated platform for managing population data in Ethiopia. The absence of such a system remains a major obstacle to accurate population analysis and evidence-based national planning.

1.3 Objectives of the Project

- ✓ The Population Count System has some goals that we want to achieve. These goals are what the Population Count System is supposed to do. We have broken down these goals into a goal and some smaller goals for the Population Count System. This helps us understand what the Population Count System is supposed to do stay focused, on the Population Count System and measure how well the Population Count System is doing.

1.3.1 General Objective

The main goal of this project is to look at the Population Count System and figure out what it needs to do. We want the Population Count System to help manage population data, in Ethiopia in a way that's accurate, secure and efficient. The Population Count System should be able to do this according to the rules that govern software requirements engineering. We will analyze the Population Count System. Write down what it needs to do so that the Population Count System works well.

1.3.2 Specific Objectives

- *The specific objectives of the project are to:*
  - ➢ Identify and analyze the population data management challenges faced by government institutions in Ethiopia
  - ➢ Elicit functional and non-functional requirements from relevant stakeholders
  - ➢ We need to create a population data model. This model should help reduce data duplication. It should also reduce inconsistency in the population data.
  - ➢ A good population data model will keep all the population data in one place. This way we can make sure that the population data is correct and the same everywhere. We will not have to worry about the population data being different in places.

- The population data model should be easy to use and understand. It should be able to store all the information we need about the population. This includes things like the number of people their ages and where they live.

We can use the population data model to make sure that our population data is accurate. We can also use it to make sure that we do not have information. This will help us to make decisions about the population.

The main goal of the population data model is to reduce data duplication and inconsistency in the population data. We want to make sure that the population data is correct and easy to use. This will help us to do a job with the population data. The population data model is very important, for this.

We need to figure out what the system needs to be able to handle population registration, updating and reporting. The system requirements should support things like adding people to the population register making changes, to existing information and creating reports based on the data. Population registration is a part of this so the system needs to be able to store and manage all the relevant information.

✧ **The system should be able to update population information**

- Population registration and updating are crucial. The system should be able to handle reporting as well so we can get a clear picture of the population.
- Address security, privacy, and access control requirements for sensitive population data.
- Ensure system scalability to accommodate population growth and administrative expansion.
- Validate and verify the documented requirements for correctness, completeness, and consistency.

1.4 Scope of the Project

The scope of this project focuses on the **requirements analysis and specification** of a **Population Count System** intended to support population data management in Ethiopia. The system is designed to provide a centralized and digital platform for recording, updating, storing, and reporting population information for authorized government users.

Within scope, the system shall support the registration of individuals and households, management of basic demographic attributes such as age, gender, and administrative location, and generation of population statistics to assist planning and decision-making at federal, regional, and local administrative levels. The system shall also address security, data integrity, and access control requirements due to the sensitive nature of population data.

The project scope is limited to the **software system requirements** and does not include physical census operations, biometric data capture, or the procurement and deployment of hardware infrastructure. Integration with external national systems, such as national identification platforms or civil registration systems, is considered outside the current scope but may be addressed in future enhancements.

This scope ensures that the project remains feasible within academic constraints while providing a clear and structured foundation for the development of a scalable population count system suitable for Ethiopia's administrative and technological context.

1.5 Significance of the Study

The importance of this study is that it helps to make population data management. In Ethiopia it is really important to have accurate and up to date information about the population. This information is necessary for planning and making policies that work. It is also needed to deliver services to people in areas. By figuring out what is needed for a Population Count System this project helps to create a digital solution to fix the problems with population data that exist right now. The Population Count System is a part of this and it will make a big difference. The goal is to have a system that works well and gives information about the population. This will help Ethiopia to make plans and provide better services to its people. The Population Count System will be very useful, in this effort.

The proposed system requirements are really helpful. They help reduce mistakes in the data. The proposed system requirements also make it easier for people to access the data. The proposed system requirements even help people in government institutions make decisions

If we have a system that keeps track of the population it can really help. The proposed system requirements and the centralized population count system can make sure everyone gets a share of resources. The proposed system requirements and the centralized population count system can also help plan services better.

The government can use the proposed system requirements and the centralized population count system to keep an eye on what's happening with the population. The proposed system requirements and the centralized population count system are useful at all levels of government from the level to the regional level, to the local level.

This study is important for schools because it shows how to use software requirements engineering principles in a real life situation. Software requirements engineering principles include figuring out what is needed looking at the needs writing down the needs and making sure the needs are correct. This study is an example for students and researchers who want to work on big information systems in countries that are still growing like Ethiopia.

The study of software requirements engineering principles is helpful for people who want to learn about software requirements engineering principles, in Ethiopia.

Overall, the study contributes to both national development efforts and academic practice by promoting the use of systematic, standards-based software engineering approaches for population data management.

1.6 Project Methodology Overview

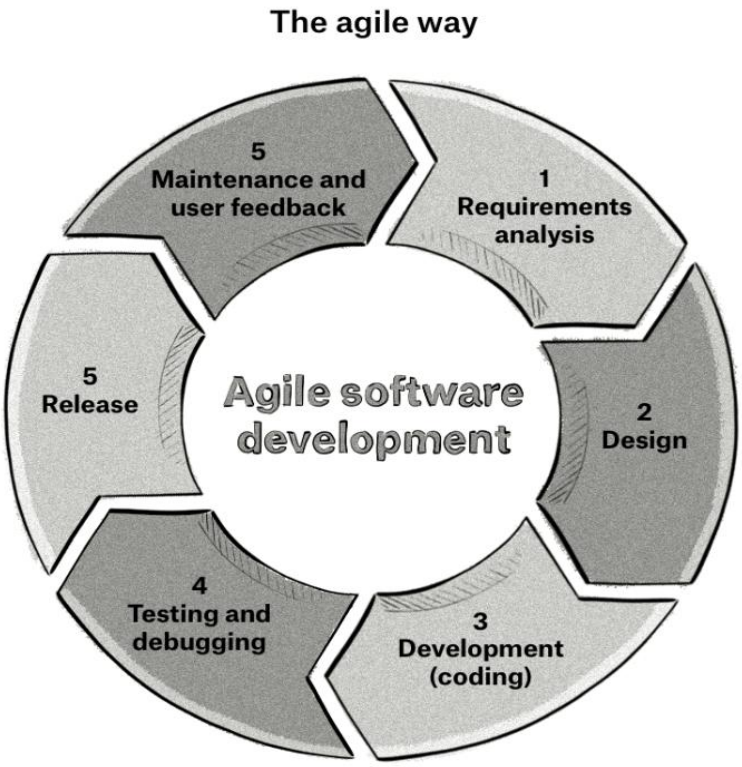
This project uses an approach to make sure the Population Count System does what it needs to do. The Population Count System has to be planned out. We need to understand what people want from the Population Count System, in Ethiopia where they have to manage a lot of population data. We have to take what these people want and turn it into requirements, for the Population Count System.

The process starts with finding out what is needed. This is where we get the information by looking at documents checking how population data is currently being used and talking to people who will be affected such as government offices and administrative authorities. We do this to figure out what the population data system needs to do and how it needs to work in the world.

This is about understanding what the population data system should be able to do and what it should be, like to use. Population data system requirements are very important here.

When we get the requirements we then do an analysis to make them clearer decide which ones are most important and fix any problems with them. We then write down these analyzed requirements in a document using standard methods, like the Software Requirements Specification so that everyone can understand them easily have all the information they need and be able to track where each requirement comes from.

Finally, requirements validation and verification activities are performed to confirm that the documented requirements accurately represent stakeholder needs and conform to software engineering standards. This methodological approach ensures that the Population Count System requirements are reliable, feasible, and suitable for Ethiopia's administrative and technological environment.



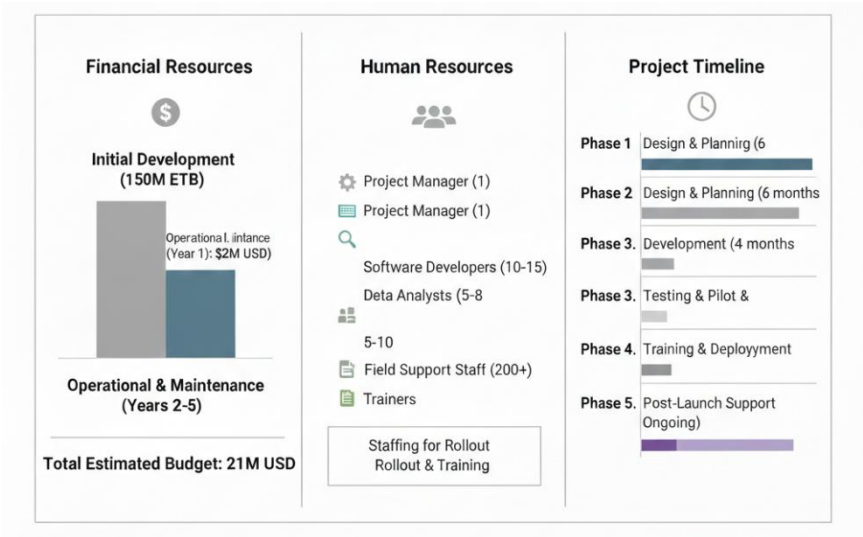
1.7 Organization of the Document

❖ This document is organized into five chapters to present the software requirements of the Population Count System in a clear and structured manner.

- **Chapter One** introduces the project by presenting the background of the study, problem statement, objectives, scope, significance, and the overall methodology used in developing the requirements.
- **Chapter Two** focuses on the analysis of the existing system. It discusses the current population data management practices, their limitations, requirements engineering process models, elicitation techniques, validation approaches, challenges, and a review of related systems.
- **Chapter Three** presents the system analysis and proposed solution. It includes stakeholder identification, business process analysis, feasibility study, system requirements, assumptions, constraints, and requirements traceability.
- **Chapter Four** describes the system modeling and design aspects. It presents architectural design, UML diagrams, data flow diagrams, database requirements, quality attributes, and the requirements traceability matrix.
- **Chapter Five** concludes the document and provides recommendations for future work and system enhancements

1.8 Budget and Development Plan

The Budget and Development Plan outlines the estimated financial resources, human resources, and timeline required to develop the Population Count System for Ethiopia. The plan ensures that the project is feasible, realistic, and can be executed within the constraints of available resources.



- ***Budget Overview***

The estimated budget for the project covers the following categories:

Category	Description	Estimated Cost (ETB)
Software & Tools	Development and testing software licenses (IDE, database, analytics tools)	5,000,000
Hardware & Infrastructure	Servers, storage devices, backup equipment	10,000,000
Human Resources	Development team, analysts, testers, documentation	30,000,000
Training & Capacity Building	Workshops for system users and administrators	2,000,000
Miscellaneous	Contingencies, printing, and administrative expenses	3,000,000
Total		**50,000,000**

Development Plan / Timeline

- The project development plan follows a **structured phased approach** aligned with software requirements engineering best practices:

PHASE	DESCRIPTION	DURATION
Requirement Analysis	Elicitation, analysis, and documentation of requirements	2-3 weeks
System Design	Database design, architecture specification, interface planning	2 weeks
Implementation	Coding of system modules and integration	3-4 weeks
Documentation	Preparation of SRS, manuals, and final project report	2 weeks
Review & Presentation	Stakeholder review and final submission	1-2 weeks
Testing	Unit testing, integration testing, verification against requirements	1 week

CHAPTER TWO

EXISTING SYSTEM ANALYSIS

2.1.1 Description of the Existing System

In Ethiopia the way they manage population data is old fashioned. They mostly use ways and some digital tools to do this job. The main way they get population data is by doing a census every ten years or so. They also use records from government offices like the Ministry of Population regional bureaus and local administrative units to get information about the population in Ethiopia. The population data, in Ethiopia is very important. They try to update it with the help of these government offices.

These systems we have now use paper forms, spreadsheets and databases that are not connected so people have to do a lot of work to collect enter and check the data. People who collect data go from house to house to get information about the people who live there the households and things, like age and gender. Then they put all this information together for each district, region and the whole country.

The system has some things that make it work in a way.

The system is made up of these things that make it special and help it do its job.

- The system has some rules that it follows

The system is what makes all of this happen. It is very important to understand how The system works.

Manual data processing is when people do all the work of collecting and entering information by hand. This means that humans are doing all the tasks, which can cause mistakes and slow things down. Manual data processing is something that relies on people to get the job done. That is why it can be prone to errors and delays, in manual data processing.

People need to know that population records are only updated when there is a census or a special survey. This means that the information about the population can become old fast.

The population records are not updated all the time during these special times like when there is a census or a special survey. This is a problem because the population information is not always current it can be outdated, which is the case, with population records.

The problem with storage is that population data is all over the place. It is stored in different institutions and regions. This means that we have the data, in lots of different places, which can cause problems. We end up with data and the information is not always consistent. Population data is stored in institutions and population data is also stored across many regions, which makes it hard to keep track of it all.

Limited accessibility is a problem. Access, to population data is not easy to get. When we need information to plan something it takes a lot of time. We have to go through a lot of administrative steps. Population data is hard to get. It is a real hassle. We waste a lot of time trying to get the population data we need.

Overall, the existing system lacks centralization, automation, and real-time updating capabilities, which hinders timely and accurate decision-making for national development and planning purposes.

2.1.2 Limitations of the Existing System

- ✧ The way they manage population data in Ethiopia is mostly done by hand or, with some computer tools. These methods have a lot of problems:

I. Data Inaccuracy and Inconsistency

Population data is gathered by hand or with digital tools, at the federal level the regional level and the local level. This can be a problem because it often means we get information, duplicate records and old data that is no longer correct. This makes the population statistics we have not very reliable. Population data is very important. We need to make sure it is accurate. Population data that is not accurate can cause a lot of issues.

II. Limited Accessibility and Timeliness

Population statistics are not ready when we need them because people have to collect and report the data by hand. This means that the people who make decisions and plans often have to wait to get the population statistics. When they finally get the data it is too late to make decisions about policies and how to use resources. The population statistics are very important, for decision-makers and planners so it is a problem that they're not available right away.

III. High Operational Costs

The old way of doing a census and keeping track of people requires a lot of people to go out and count everyone. It also needs a lot of help with things, like transportation and equipment.. It costs a lot of money. This makes it hard to keep track of population data all the time. The traditional census is just not easy to do because it is so expensive.

IV. Poor Scalability

The existing systems are not good at handling the population growth that happens fast or the migration trends especially in the cities where the population changes a lot. This is a problem because the system has a hard time giving complete coverage to all the different areas that need to be taken care of. The existing systems just cannot keep up with the population growth and the migration trends. That is why they struggle to provide comprehensive coverage for all the administrative units, like the existing systems are supposed to do for the administrative units.

V. Limited Data Security and Privacy

We have to deal with paper records and computer systems that are not connected. These systems do not have security to control who can access the information. This means that someone who is not supposed to see the information might be able to get to it. We could lose the information or someone might use the information about people in a bad way. The problem is, with the paper records and the computer systems that are not connected. They are not safe. That is why we have to worry about people getting to the information when they are not supposed to.

VI. Difficulty in Integration

The absence of a centralized data platform prevents seamless integration with other government systems such as civil registration, health, and education databases, limiting the usefulness of population data for cross-sector planning.

2.2 Requirements Engineering Process Models

The requirements engineering process defines the structured activities through which stakeholder needs are identified, analyzed, documented, and validated to produce a reliable software requirements specification. For the **Population Count System** in Ethiopia, applying a formal requirements engineering process is essential to ensure that the system addresses local population data challenges, government policies, and administrative realities.

Several process models can be used in requirements engineering, each providing a structured approach to gather and refine requirements:

Waterfall Model for Requirements Engineering

This sequential model defines distinct stages: **elicitation → analysis → specification → validation**.

- ***Strengths:***
  - ✓ Clear structure and documentation at each stage
  - ✓ Easy to track progress
- ***Limitations in Ethiopian context:***
  - ✓ Inflexible to rapidly changing requirements, such as sudden policy changes or migration data updates
  - ✓ Requires early and complete stakeholder availability, which may be challenging across regional administrations

Iterative or Incremental Model

Requirements are developed in cycles, with partial implementations reviewed and refined at each iteration.

- ***Advantages:***

  - ✓ Accommodates evolving requirements and stakeholder feedback
  - ✓ Supports early identification of data and process gaps
  - ✓ Aligns well with Ethiopia's dynamic population tracking needs

- ***Application:***

  - ✓ Stakeholder feedback from federal, regional, and local offices can be incorporated in successive iterations of the requirement specification

Prototyping Model

Focuses on building a working prototype of the system based on preliminary requirements to gather feedback.

- ***Strengths:***

  - ✓ Provides concrete visualization of the Population Count System
  - ✓ Improves communication with non-technical stakeholders
  - ✓ Useful in Ethiopia, where some local offices may have limited understanding of digital systems

- ***Limitations:***

  - ✓ May lead to incomplete or evolving requirements if feedback is not structured

Agile or Incremental Requirements Engineering

Emphasizes close collaboration with stakeholders, frequent review, and flexible adaptation of requirements.

- ***Strengths:***

  - ✓ Encourages active participation from government users and data administrators
  - ✓ Supports continuous refinement of requirements in response to real-world population dynamics

- ***Challenges:***

  - ✓ Requires consistent stakeholder involvement and communication channels

Recommended Approach for the Population Count System

Considering the Ethiopian context — including **regional administration structures, diverse stakeholder availability, and dynamic population changes** — a **hybrid approach combining iterative and prototyping models** is recommended:

- Initial **elicitation** through interviews, document review, and observation
- **Prototype** development for visual feedback on population data management workflows
- Iterative **analysis and refinement** to accommodate stakeholder feedback across federal, regional, and local offices
- **Validation and verification** to ensure completeness, accuracy, and compliance with national data policies

This approach ensures that requirements are both **accurate and adaptable**, producing a robust SRS suitable for implementation in Ethiopia.

2.3 Requirements Elicitation Techniques

The Population Count System in Ethiopia needs to know what people want. We have to find out what the government and other important groups need from this system. For the Population Count System in Ethiopia we need to ask the questions so that the system is useful for the government, regional administrations and data officers. The Population Count System in Ethiopia has to work in the world so we need to use good techniques to figure out what people really need from the Population Count System, in Ethiopia.

- These are the techniques that people think you should try:

- **Interviews**

  Conducted with key stakeholders such as government officials, regional data administrators, and local census officers.

  Purpose: To gather detailed information about current population data processes, challenges, and expectations for the new system.

When we think about Ethiopia we have to consider a things. The people involved in our project are spread out across regions. So we might need to do interviews in person and also have some interviews online. It really depends on how easy it's for people to get to where we need them to be. We will use a combination of face-to-face interviews and remote interviews, for Ethiopia.

- **Questionnaires and Surveys**

  Used to collect inormation from a large group of stakeholders efficiently.
  Useful for standardizing input on population attributes, reporting requirements, and system expectations.

  When we are talking about Ethiopia we need to think about the languages that people speak there. We have to make sure that our questionnaires are translated into languages like Amharic and Oromiffa.
  This is important for the officers in Ethiopia so they can understand the questions and do their jobs properly. We are talking about Ethiopia and the people who work there so we need to use languages that're relevant, to Ethiopia.

- **Document Analysis**

  Review of existing population records, census reports, administrative forms, and statistical summaries.
  Purpose: To understand current data structures, gaps, and inconsistencies.
  When we think about Ethiopia we have to remember that a lot of their records are still, on paper or partly computerized. So when we analyze this information we have to keep in mind that some of the data might be missing or not entirely accurate. This is something we have to consider when looking at Ethiopias records because Ethiopias data can be incomplete or inconsistent.

- **Workshops and Focus Groups**

  We have meetings with the people involved in a project, like stakeholders to talk about what we need to do. We want to make sure everyone knows what to expect from each other. Sometimes people disagree so we use these meetings to work through our differences and find a solution that works for everyone. We call these meetings sessions with stakeholders.

  Purpose: To achieve consensus on system functionality and data requirements.

When we think about Ethiopia we have to consider a things. Regional representation is really important for Ethiopia. This is because we want to make sure that every part of Ethiopia is included, across all the different administrative zones, in Ethiopia.

- **Observation**

Direct observation of current population data collection and management practices.
Purpose: To identify inefficiencies, repetitive tasks, and manual data entry errors.

When we think about Ethiopia we should consider something. If we go to offices and meet the people who are doing the census we might find out some problems that are not written down in the official reports. This is because field visits, to these offices and census teams can show us the real challenges they are facing every day in Ethiopia.

- **Prototyping**

Development of preliminary system mock-ups to gather feedback.
The purpose of this is to let people who are interested in the system see what it looks like make sure the functional requirements are what we really need and check that the system is easy to use. This is, about the system and how it works so we want to make sure the system is something that people can actually use. We need to look at the system and make changes to the system if necessary. The system is important so we want to get it right.

Ethiopia-specific consideration: Useful for stakeholders with limited experience with digital systems.

Recommended Approach for Ethiopia

For the Population Count System I think it is an idea to use a mix of different methods. We should do interviews with people look at documents and try out ideas with prototyping. This way we can get an understanding of the Population Count System. The Population Count System will benefit from this approach because it will give us more complete information.

We will have interviews with federal officers and key regional officers to find out what the primary requirements of the federal officers and the regional officers really are. This will help us understand what the federal officers and the regional officers need.

- The interviews will be with federal officers
- The interviews will also be with regional officers

The goal of these interviews, with the federal officers and the regional officers is to capture the primary requirements of the federal officers and the regional officers.

Document analysis to validate existing data practices and identify gaps.

When we are making a system we need to show people how it works. We do this by creating a prototype. A prototype is like a model of the system. It helps us to visually communicate what the system can do. This is really important for stakeholders who do not know much about systems. We use the prototype to show them how the system functions. This way stakeholders can understand the system better even if they are not familiar, with systems. We are prototyping to help stakeholders see what the system can do.
Questionnaires/focus groups for additional input and consensus across regions
This approach ensures that the elicited requirements are accurate, complete, and feasible for implementation in the Ethiopian administrative context.

2.5 Requirements Analysis and Negotiation

The Population Count System in Ethiopia needs to be planned. This is an important step in Software Requirements Engineering. We need to make sure we understand what the Population Count System in Ethiopia needs. This means we have to take all the information we collected about the Population Count System in Ethiopia and make it clear and easy to understand. We also have to make sure we have all the information we need for the Population Count System in Ethiopia. The Population Count System in Ethiopia has to work and do what it is supposed to do. So we have to take all the information about the Population Count System in Ethiopia and turn it into instructions that the software developers can use to build the Population Count System, in Ethiopia.

When we are looking at the requirements we can put them into groups. These groups are:

Functional Requirements

The system will do a things. It will register individuals and households. The system will also update information for individuals and households.

The system will generate population reports.
The system will manage user access, to the system. This means the system will control what each user can do in the system. The system will manage user access.

Non-functional Requirements

We need to think about what makes a system. The system should be able to do its job which is performance. It should also be able to handle a lot of work which's scalability. The system has to be secure so people who are not supposed to use it cannot get in. We also need to make sure the system keeps our data safe this is called data integrity. Lastly the system should be easy to use, which is usability. The system should have performance, scalability, security, data integrity and usability.

<u>Constraints</u>

Include limitations due to Ethiopian ICT infrastructure, regional administrative processes, budget constraints, and legal requirements regarding data privacy.

Negotiation is really important. It comes after we do our analysis. It is about finding a way to deal with demands that do not agree with each other and figuring out which ones are more important so we can meet the needs of the people who have a stake in it.

In the context of Ethiopia negotiation usually happens between groups, such, as:

- ✓ Federal government agencies
- ✓ Regional bureaus of statistics
- ✓ Local administrative offices
- ✓ Negotiation addresses conflicts such as:
- ✓ Resource constraints vs. system functionality
- ✓ Security and privacy requirements vs. accessibility

Data completeness vs. practical limitations in remote regions
The Population Count System needs to make sure that everyone involved agrees on what's possible and what is real. This means talking things over so that the Population Count System does not get held up.
If people do not understand what is required it can cause problems and the Population Count System may not work properly. This is especially important for something big, as the Population Count System that is used all over the country.

Things you do in Requirements Analysis and Negotiation include:

- ✓ Reviewing elicited requirements for clarity and completeness
- ✓ Identifying conflicting or ambiguous requirements
- ✓ Prioritizing requirements based on criticality and feasibility
- ✓ Facilitating stakeholder meetings to resolve conflicts
- ✓ Documenting agreed-upon requirements for inclusion in the SRS

2.6 Requirements Specification Standards

Requirements specification standards provide guidelines and best practices for documenting software requirements in a clear, consistent, and verifiable manner. Adhering to recognized standards ensures that the requirements of the **Population Count System** are understandable to all stakeholders, including analysts, developers, testers, and government decision makers in Ethiopia.

For this project, the requirements are documented using the principles of the **Software Requirements Specification (SRS)** standard, such as those defined by **IEEE standards**.

These standards emphasize completeness, correctness, consistency, and traceability of requirements throughout the system development lifecycle.

According to these standards, each requirement shall be:

- **Clear and unambiguous** – stated in precise language to avoid multiple interpretations
- **Complete** – fully describe the required system behavior
- **Consistent** – not conflict with other specified requirements
- **Verifiable** – testable through inspection, demonstration, or analysis
- **Traceable** – linked to stakeholder needs and system objectives
- **Feasible** – achievable within technical, operational, and organizational constraints

In the context of Ethiopia, applying requirements specification standards is especially important due to the involvement of multiple administrative levels and institutions. Standardized documentation ensures uniform understanding of system requirements across federal, regional, and local government offices, reducing miscommunication and implementation risks.

The Population Count System requirements are organized into functional and non-functional categories and documented using structured language, standardized formatting, and consistent terminology. This standardized approach supports effective validation, future system maintenance, and potential system expansion.

2.7 Requirements Validation and Verification

Requirements validation and verification are essential activities in Software Requirements Engineering that ensure the documented requirements accurately represent stakeholder needs and are suitable for system development. For the **Population Count System** in Ethiopia, these activities are critical due to the system's nationwide impact and the sensitivity of population data.

Requirements validation focuses on confirming that the specified requirements reflect the actual needs and expectations of stakeholders. It answers the question: *"Are we building the right system?"* Validation ensures that the Population Count System

requirements align with government policies, administrative procedures, and real-world population data management practices in Ethiopia.

Requirements verification, on the other hand, ensures that the requirements are correctly documented according to specification standards. It addresses the question: *"Are we building the system right?"* Verification checks that the requirements are complete, consistent, unambiguous, and conform to established SRS standards.

Validation and verification activities for this project include:

- ✓ **Requirements reviews** involving domain experts and stakeholders from federal, regional, and local administrative levels
- ✓ **Checklist-based inspections** to ensure compliance with SRS standards
- ✓ **Scenario analysis and use case reviews** to confirm realistic system behavior
- ✓ **Stakeholder feedback sessions** to identify missing or incorrect requirements

Applying validation and verification reduces the risk of misunderstandings, rework, and system failure. For Ethiopia's Population Count System, these activities help ensure that the final system supports accurate data collection, secure data handling, and effective decision-making across all administrative levels.

ASPECT	REQUIREMENTS VALIDATION	REQUIREMENTS VERIVICATION
Purpose	Ensures the right requirements are defined	Ensures requirements are correctly documented
Key Question	Are we building the right system?	Are we building the system right?
Focus	Stakeholder needs and expectations	Requirement quality and compliance
Performed By	Stakeholders, domain experts, analysts	Analysts, reviewers, quality assurance team
Timing	After requirements specification	Throughout requirements documentation
Outcome	Confirms requirement relevance	Confirms requirement correctness

2.8 Challenges in Software Requirements Engineering

Software Requirements Engineering for the Population Count System in Ethiopia is really tough. The Population Count System in Ethiopia has a lot of issues that can hurt its quality and success. This is because the Population Count System in Ethiopia is used over the country. Many people are involved with the Population Count System in Ethiopia. They all have different ideas. The Population Count System in Ethiopia also has to work in different places, with different rules.

One big problem is that there are different stakeholders and they all need to talk to each other. Managing population data involves a lot of people from the level the regional level and the local level. Each of these groups has its priorities, its own way of understanding technical things and its own needs when it comes to getting things done. When these groups do not understand each other or when they do not communicate well it can lead to problems, like missing information or requirements that do not match up with each other. Population data management is really hard because of all these stakeholders and their different needs.

Changing requirements are a problem. The needs of the population and the rules that govern a place can change. The lines that separate one area from another can also shift. What people need to know from the data can be different, over time. It is hard to pin down what is really needed when everything is moving fast. This means we have to keep looking at the requirements of the population-related policies the administrative boundaries and the data reporting needs all the time to make sure we get it right. The requirements of the population-related policies and the administrative boundaries can be very tricky to capture.

In some parts of Ethiopia the technical infrastructure is not very good. This is another problem. The fact that the network is not always available and that we have levels of hardware and technical expertise in different areas can make it hard to decide what the system really needs. The system requirements are affected by these variations in network availability and technical expertise across Ethiopia. This affects what we can do with the system, in Ethiopia.

Data sensitivity and privacy concerns are a problem. The information about people, in the population data is very personal so it needs to be kept safe and secure. We need to make sure that people can use the data when they need to. We also need to protect peoples privacy. This is very hard to do. We have to be very careful when we are figuring out what we need to do with the population data and the data sensitivity and privacy concerns. We have to balance the data accessibility with the privacy protection of the population data. This is not easy to do.

We have a problem when the paperwork for what we do every day is not complete or is hard to understand. This can make it tough to figure out what we really need. Sometimes the way we handle information about people is not written down properly so it is hard to know how things are done now. This is especially true for things, like handling population data, where the procedures might not be formal or might not be written down all.

Addressing these challenges requires a structured requirements engineering approach, active stakeholder involvement, clear documentation standards, and continuous validation throughout the project lifecycle.

2.9 Review of Related Systems

The review of systems looks at how other countries manage and count people. This review checks out what systems are already in place. It helps figure out what works well what features are useful and what does not work well. All of this information is helpful for the Population Count System that is being planned for Ethiopia. The Population Count System, for Ethiopia can learn from what other countriesre doing.

Many countries are now using systems to keep track of their population and manage census data. This helps them get more accurate and up to date information about the people living in their country.

These digital population registration and census management systems usually have a place to store all the data and they get updated on a regular basis.

The digital population registration and census management systems also make it easy to create reports and control who can see the information so only the people who are supposed to have access, to the population registration and census management systems can use them.

The National Census Management Systems are really helpful. They make it easier to plan the census and collect data. They also make it easier to report the results. This is better, than doing everything by hand.. These systems usually collect data at certain times. This means they do not always show the population. The National Census Management Systems have this problem because they do not collect data all the time. The National Census Management Systems are good. They need to be able to collect data more often to show the real population.

The Civil Registration and Vital Statistics System is another system that is related. This system is used in countries to record important life events like when people are born when people die and when people get married. The Civil Registration and Vital Statistics System gives us updates all the time.. It has some limits. It only records important life events, like births and deaths. It does not always show how people are moving from one place to another or what is happening at the household level. This information is necessary to understand the population. The Civil Registration and Vital Statistics System is useful. It does not give us all the information we need to do a complete analysis of the population.

Some countries have National Population Registry Systems in place. These systems keep track of peoples information. It is always up to date. This information is linked to the identification number that each person has. National Population Registry Systems are good because they make sure the information is correct and they can work well with systems.. National Population Registry Systems need a few things to work properly. They need computer systems and laws to support them. They also need everyone in the country to work together. This can be hard to do in some countries that are still developing. National Population Registry Systems can be a challenge, for these countries.

In Ethiopia the management of population data is done mainly by having censuses from time to time and by keeping administrative records at different institutions.

The problem with these systems is that they are not fully connected they do not get updated away and they are not all in one place, which causes the population data to be incorrect and makes it hard to get the information when it is needed.

Population data management, in Ethiopia is really important. It is affected by these issues with the systems used for population data management.

The review of these related systems highlights the importance of designing a Population Count System that balances functionality, feasibility, and scalability. Lessons learned from existing systems inform the requirements of the proposed system, ensuring it addresses Ethiopia's specific administrative, technological, and operational conditions.

CHAPTER THREE

SYSTEM ANALYSIS

3.1 Overview of Software Requirements Engineering

Software Requirements Engineering is an organized way to make sure we get the software requirements right. It is about finding out what the software needs to do looking at it closely saying exactly what it should do checking that it is correct and managing all of this from start to finish. Software Requirements Engineering helps people who're interested in the software and the people building it understand what the software should be able to do. This understanding is very important, for building a software system.

The Population Count System is very important. For this system to work properly we need to make sure we get the software requirements right. This means we have to say what the system should do and what it should not do. We also have to make sure these requirements are consistent complete and can be checked before we start designing and building the system.

The Population Count System deals with population data which's very sensitive. This data is also crucial for planning at the level. So if we make mistakes or are not clear, about what we want it can cause problems. These problems can be operational, legal or affect the economy and society. The Population Count System needs to be done to avoid these issues.

In Ethiopia people need to think about the problems that come with figuring out what is required. This includes dealing with government offices some places having better technology than others and lots of people from different government levels being involved.

This makes it more likely that people will want things that we will not have all the information we need and that people will not communicate properly.

Using a planned approach to figuring out what is required can help reduce these problems.

It does this by giving us a step by step way to talk to the people involved and to make sure we get the requirements right. Requirements engineering, in Ethiopia is very important. Requirements engineering helps with this.

The software requirements engineering process that we are using for this project has a main things that we need to do.

The software requirements engineering process for this project includes some steps.

We have to follow the software requirements engineering process, for this project. It has some key parts.

These are the things we have to do for the software requirements engineering process of this project:

To figure out what people really need we have to do something called Requirements Elicitation. This is when we carefully find out what stakeholders want by looking at documents talking to them and watching how they currently manage data. We do this to understand the needs of the stakeholders. Requirements Elicitation is important, for understanding the needs of the stakeholders through document analysis, interviews and observation of existing population data management practices.

Requirements Analysis: Evaluation and refinement of elicited requirements to detect inconsistencies, redundancies, and feasibility issues, while prioritizing requirements based on system objectives and constraints.

Requirements Specification: Formal documentation of requirements using standard Software Requirements Specification (SRS) formats, ensuring clarity, precision, and traceability.

Requirements Validation and Verification is a part of making sure the software is what the people who will use it really need. We look at the requirements that are written down to see if they are correct and if they follow the rules of software engineering.

The main goal of Requirements Validation and Verification is to make sure that the requirements accurately reflect what the stakeholders want and need.

We do this to ensure that Requirements Validation and Verification is done properly and that the software meets the requirements.

By applying a rigorous software requirements engineering approach, this project aims to reduce development risks, improve system quality, and ensure that the Population Count System is aligned with Ethiopia's administrative framework, data governance needs, and long-term development objectives.

3.2 Proposed System Overview

The new Population Count System is a computer program that will help Ethiopia with its population data. This Population Count System is supposed to fix the problems that exist with the way population information is handled now. The Population Count System wants to give the government an trustworthy way to manage population information. This will be useful, for the government, the regional government and the local government. The Population Count System will be a place where all population information can be stored and used.

The new system will help people register and update information about the population digitally. It will. Give back this information when it is needed. This system is meant to replace the ways of doing things that were not connected and were done by hand. The old ways were not very good because they did not always have the information and it was hard to get to. The new system will make sure that all the information is correct and easy to get to.

By putting all the population records in one place the system will stop information from being repeated over and over. This will make it easier for people to work together to get the information they need to make plans and decisions, about the population. The population data will be easy to get to and use for planning and making decisions.

The Population Count System is made for use in Ethiopia. It will work with the countrys administration and technology. This system will have to deal with levels of administration and it will help users who have permission to use it. These users will have jobs and they will be able to do different things on the system.

The Population Count System will also think about the fact that some areas in Ethiopia have technology and more people who know how to use computers, than other areas. The Population Count System has to work in Ethiopia.

The proposed system is supposed to do a few things. It needs to help manage population records make sure users are who they say they are and give them access to the things they need to see. The system also needs to be able to make reports, about the population.

The system has to be secure keep the data safe be able to handle a lot of people using it and be easy to use. This will help make sure the population record management system and the user authentication and authorization system work well for a time and do not fail. The system has to be reliable. This is very important for the population reports and the population record management system.

Overall, the proposed system serves as a foundational platform for modern population data management in Ethiopia, providing a requirements-driven solution that aligns with national planning needs and software engineering best practices.

3.3 Stakeholder Identification and Analysis

The Population Count System involves multiple stakeholders due to its role in managing sensitive population data across different administrative levels in Ethiopia. Identifying these stakeholders is essential to ensure that system requirements reflect real operational needs and responsibilities.

The primary stakeholders of the Population Count System are **government institutions responsible for population administration and planning**. At the federal level, these institutions require aggregated and accurate population data to support national policy formulation, development planning, and resource allocation. Their requirements focus on data accuracy, consistency, reporting capabilities, and system reliability.

Regional, zonal, woreda, and kebele administrations are key operational stakeholders. These entities are responsible for registering population data, updating demographic information, and maintaining local records. The system must support their daily activities by providing efficient data entry, update mechanisms, and access to localized population reports that reflect administrative hierarchies.

System administrators are responsible for managing system operations, including user accounts, access permissions, and system configuration. The Population Count System must support administrative functions such as role-based access control, system monitoring, and secure management of population data.

Data registration officers interact directly with the system to record individual and household information. Their requirements emphasize ease of use, data validation, error handling, and performance to ensure accurate and efficient population data entry.

Government planners and decision makers use the system outputs for planning and analysis. They require timely access to population statistics, summary reports, and demographic breakdowns to support decision-making in sectors such as education, healthcare, and infrastructure development.

Citizens are indirect stakeholders whose personal data is stored and managed by the system. Although they do not interact with the system directly, their interests are reflected in requirements related to data privacy, confidentiality, and secure handling of personal information.

The identification and analysis of these stakeholders ensure that the Population Count System requirements address operational, administrative, and policy-related needs while aligning with Ethiopia's population management structure.

3.4 Business Process Analysis

The Population Count System is a tool that helps with the work of managing population data, in Ethiopia. It is used across the country following the same structure that the government uses to divide up the land. The Population Count System is made to work with the systems that the government already has in place. This helps to make things more efficient, accurate. It also helps different institutions work together better with the Population Count System.

The main business process starts with people registering in their area. The people in charge of registration write down information about each person and their family into the system. They do this based on the borders of the area.

The system checks the information that is entered to make sure it is correct and has everything it needs before it is saved in the database. The population registration is an important part of this process. The system is used to store information, about the population at the administrative level.

The second main thing the system does is update population data. When something happens like a birth or a death or people move to a place the people in charge put this information into the system. The system makes sure that all these updates can be tracked and that the old data is still good so we can see what the population is like now and what it was, like before.

Population data management and verification is an important part of our business. We have people who supervise and they check the records that other people send in. They do this to make sure everything is correct and the same. The system we use helps with this by letting us control who can see the information by having a way to approve things and, by keeping a record of everything that happens to the data. Population data management and verification is something we take seriously.

The system is also able to bring population data and create reports. The population data is put together based on how the country's divided up so we can make reports for the whole country for a region for a zone for a woreda or for a kebele.

These population reports help with planning keeping an eye on things and making decisions, about the population data. We use the population data to make these reports which're useful for people who need to know about the population data.

The system makes it easy for people to get into it and for the administrators to manage it. The administrators of the system are, in charge of what the users can do what they are allowed to see. How the system is set up. They do this so that only the people who are supposed to can look at the population data and make changes to it.

The system is used to keep the population data safe. The administrators make sure that only authorized users can get to the population data and change it.

By supporting these business processes, the Population Count System streamlines population data management, reduces reliance on manual procedures, and enhances coordination among government institutions in Ethiopia.

3.5 Feasibility Study

The Population Count System is being checked to see if it can really work in Ethiopia with the technology and money they have now. This check shows that the Population Count System project is an idea and can be used all over the country to manage information, about the population. The Population Count System can be. Used in Ethiopia because it fits with what they have and what they can afford.

3.5.1 Technical Feasibility

The Population Count System can actually work in Ethiopia because they already have the technology. The government uses computers to manage information. They have the basic technology in place, at the federal and regional levels. This technology is also being used more and more at offices. The Population Count System is something that can be done with the systems they already have in Ethiopia.

The system is made to work with the computer and internet we already have. It does not need machines or equipment. The system can use the computer and the usual ways we store and share information. The system is flexible. Can work with different types of internet connections. It can store data in one place and control who gets to see it. The system works with the computer and database and networking technologies that are easy to find.

The Population Count System does not need a lot of skills to run and maintain. The government IT departments already have the people who can do this job. If they do not have these skills they can learn them with some training. So the technical things that the Population Count System needs are easy to do with the technology we already have. The Population Count System is something that we can handle with what we have.

3.5.2 Economic Feasibility

The new Population Count System is an idea because it saves money in the long run. It costs a lot to collect population data the way. We need to pay for people to do the work vehicles to get them around and paper to write down all the information. The Population Count System is better because it reduces these costs. Collecting population data is easier, with the Population Count System.

The population data management system is made better by using computers. This means that people do not have to collect the data over and over again. The system also helps to reduce the money that is spent on running it. This makes the whole process work smoothly.

The system is about creating and updating software. It does not need a lot of money to be spent on computer equipment or other things, like that.

The good things about having the population numbers at the right time for planning and giving out resources are more important than the money it takes to set up and start the Population Count System. This is why the Population Count System is a way for Ethiopia to manage its population needs without spending too much money. The Population Count System is really helpful, for Ethiopias population management needs.

3.5.3 Operational Feasibility

The Population Count System can actually work well in Ethiopia. It fits in with the way things are already done by the government.

The system works with the people who're in charge of managing the population at the federal level the regional level and the local level. This means that the people, in charge can use the Population Count System as part of their work without having to make a lot of big changes. The Population Count System is something that institutions can use easily.

The system is made to work with role-based access. This means that users of the system do things according to their jobs. The system is designed to be easy to use. It has structured workflows. This helps users who have levels of technical experience.

The system supports users in this way so they can do their jobs easily. The system is really about making sure that users can do what they need to do based on their roles, in the company.

With appropriate user training and administrative support, the system can be smoothly adopted and operated by government offices. This ensures effective system utilization and long-term sustainability.

3.6 The Requirements Elicitation Techniques That Were Used

These are the methods the team used to gather the requirements for the project. The requirements elicitation techniques that were used include things like talking to people and finding out what they need from the system. The requirements elicitation techniques that were used are very important because they help the team understand what the system should be able to do.

Some of the requirements elicitation techniques that were used are:

- ✓ Asking people questions to find out what they want
- ✓ Looking at what other systems do to get ideas, for the system
- ✓ The requirements elicitation techniques that were used also involved checking out what people are saying about systems to get a better understanding of what is needed.

The Population Count System needs to be done so we used many ways to figure out what is required. We did this to make sure that what we write down is what the people of Ethiopia really need to manage population data. We picked these methods because we have many different people involved and the work of managing population data, in Ethiopia is complicated. The Population Count System has to work for everyone that is why we used many methods to find out what the Population Count System needs.

We looked at documents to see how the government manages people and what rules they follow. This helped us find out what information we need to keep track of, how to report it and what laws the system has to follow. We used this information to understand what the population management procedures are and what records and policy guidelines the government institutions use.

We talked to the people who are involved in population registration to find out what they need. These people were from offices and government departments that deal with population planning. We wanted to know about the problems they face. What they expect from the system.

The interviews with these stakeholders were very helpful, in understanding what is required for population registration, updating data verifying information and making reports. Population registration is a task and we need to make sure we get it right.

We looked at how thingsre done now to see how population data is collected and managed by local administrators. This helped us find problems, things that people have to do by hand and places where we can use computers to make things easier. We want to know how population data is handled at the level so we can make it better. Population data is very important. We need to make sure it is collected and managed in a good way.

We had meetings with the people who are affected by this project, which we call stakeholder consultation sessions to make sure the things we need to do are correct and to make some changes to them. The things people told us during these meetings really helped us to understand what people are supposed to do what they need to get into the system and how they should handle the information at levels of administration like at the top or at the bottom and this is, about the data handling practices of the stakeholders.

By combining these elicitation techniques, the project ensures that the Population Count System requirements are comprehensive, realistic, and aligned with Ethiopia's population management environment.

3.7 System Requirements

This part is, about what the Population Count System can do and what it needs to follow. The Population Count System has to do things and it has to do them in a certain way. The people who will use the Population Count System have said what they need. This is based on how Ethiopia manages its population and how the government works. The Population Count System has to work with these rules so it can do its job properly.

3.7.1 Functional Requirements

The Population Count System needs to do things. It has to provide services and operations. These services and operations are what the Population Count System is supposed to do. The Population Count System has to make sure it can do all these things.

FR1: User Authentication and Authorization

The system will need all users to log in with an username and password. This means that every single user of the system has to do this. The system needs to make sure that people are who they say they are so it will ask for credentials, from all users.

The system needs to let people do things based on their role. This means administrators and registration officers and planners will all have things they can do. The system has to support this so that each group of users can only do what they are supposed to do. The system will have role-based access control, for these user categories, including administrators and registration officers and planners.

The system will not let everyone see the population data. It will only let people see this information if they have the user role. The system is set up to keep the population data safe by allowing people with certain user roles to access it. The population data is for people, with the correct user roles.

FR2: Population Registration

The system will let people who are allowed to use it add population records one, by one. This means that authorized users can put in population records. The system is made to help authorized users register these population records.

The system needs to help people register the population in each household. The system has to support this kind of registration for every household. This means the system will keep track of the population based on where people live which is their household. The household-based population registration is what the system shall support.

The system will get the information about people like their name, how old they are if they are a man or a woman and where they live for official purposes.

The system needs to collect this data, like name, age, gender and administrative location to work properly. The demographic data, including name, age, gender and administrative location is necessary, for the system.

The system needs to check the information that people enter before it saves that information. This way the system can make sure it does not have any wrong information. The system has to do this to prevent problems with the records of the system. The system will validate the data that people enter to keep the records of the system correct and complete.

FR3: Population Data Update

The system needs to let people who are allowed to make changes update the population records. This means that authorized users can do this with the population records. The population records have to be updated by these authorized users.

The system will help us keep track of things that happen to people like when someone's born, when someone dies and when people move to a new place. We need to record these population events, like birth, death and migration so we have the information.

The system has to keep the data consistent when it is being updated. The system will make sure that the data stays consistent during these updates. This means the system will always keep the data consistent.

The system will keep all the data so we can look back at it later for auditing and when we need to remember something. The system is really important for keeping data safe. We need the system to keep data, for auditing and reference purposes.

FR4: Population Data Verification

The system needs to help people, in charge check the population records. This means that people who are supervising will be able to verify the population records using the system. The system is going to make it possible for supervisory users to check population records.

The system will let people say yes or no to the records that are sent in. This means the system will let people approve or reject the submitted records. The goal is to make it easy for people to approve or reject these submitted records.

The system needs to keep track of what people do when they verify things so we can see who did what. The system will record all the verification actions, for this reason. This way the system will have a record of the verification actions.

FR5: Population Data Management

The system will keep all the information about people, in one place. This main place is a collection of data called a database. The database is where the system stores population data. The population data is stored in this centralized database.

The system needs to make sure that it does not have population records. It has to stop this from happening so that the population recordsre always accurate. The system must prevent population records at all times.

The system needs to help us find and get population records when we look for things. We should be able to search for population records using the criteria that we have defined for the population records. This way we can easily find the population records we need.

FR6: Reporting and Data Aggregation

The system will make reports, about the population based on levels. The system will use these levels to create the population reports. The population reports are very important. The system will generate them.

The system will have to provide information about the population. This information will include a version and a longer version with a lot of details, about the population. The system needs to support these two kinds of population statistics which're the summary and the detailed population statistics.

The system needs to let people who are allowed to use it get the population reports out of it. The population reports are what we are talking about here. The system has to make it possible for authorized users to export these population reports.

FR7: Administrative Management

The system needs to let people in charge, like system administrators take care of user accounts. This means they have to be able to manage user accounts. The system has to give system administrators the ability to do this.

The system will let us make user roles change existing user roles and stop using old user roles. We can create user roles modify user roles and deactivate user roles when we do not need them anymore.

The system is supposed to give us the ability to see the configuration of the system and also to maintain the system. We need to be able to configure the system and perform maintenance tasks on the system. This is important, for the system.

FR8: Administrative Hierarchy Management

The system needs to help organize population information for Ethiopia. This is based on how Ethiopia's divided up which includes federal regional, zonal, woreda and kebele levels. The system will use these levels to organize the population data, for Ethiopia.

The system will make sure that each population record is linked to an administrative unit. The population record is going to be connected to the unit. This way the population record and the administrative unit are tied together.

The system will limit what people can see based on their job and what they are allowed to do. This means that the system will control what information the administrative level and the user role can view. The administrative level and the user role will only be able to see what the system allows them to see.

FR9: Data Validation and Integrity Control

The system will make sure that people fill out all the information when they register. The system has to make sure that mandatory fields are completed during the registration process. The system is going to check the fields every time someone registers.

The system needs to check that the demographic data is in the format and, within the correct ranges. The demographic data has to be validated by the system to make sure it is correct. The system will check the data formats and the demographic data ranges.

The system will stop people from sending in records that're not complete or do not make sense. The system has to make sure that all records are complete and logical so it will prevent submission of records and also prevent submission of records that are logically inconsistent. The system is, in place to ensure that records are good and it does this by preventing the submission of records like incomplete records or logically inconsistent records.

The system has to make sure that the data is correct and safe when we create it when we update it and when we delete the data. We need to protect the data integrity when we do these things to the data. The data integrity is very important during these operations, like creation and update and deletion of the data.

FR10: Audit Trail and Activity Logging

The system will write down every thing that users do. This includes when users make data update old data check to see if the data is correct and delete data. The system has to keep track of all these actions that users perform, like data creation, updates, verification and deletion.

The system will keep a record of what happens. This record is called an audit log. The audit log will have the time and date of when something was done which is called a timestamp. It will also have the name of the person who did it which is called a user identifier. The system will store these audit logs so we can look at them later. The audit logs are important because they help us keep track of what the system's doing and who is using the system.

The system needs to let people in charge like authorized administrators look at the audit logs. This is so they can keep an eye on things and make sure everything is okay. The audit logs are important, for monitoring and accountability of the system. The authorized administrators will use these logs to check what is going on with the system.

FR11: Duplicate Detection and Record Matching

The system will find people who might be listed more than once in the population records. It does this by looking at things that are defined as the matching criteria for population records. The system checks the population records to see if there are any population records based on the defined matching criteria, for population records.

The system will tell the people who are allowed to know when it finds records that're the same as other records. The system will do this so that authorized users are aware of records. When the system finds records it will notify the authorized users, about the duplicate records.

The system will help us deal with records in a controlled way. It has to support this process so we can fix records properly. The system has to do this to make sure the duplicate records do not cause any problems.

- ✓ The system will find records
- ✓ The system will help us decide what to do with the duplicate records
- ✓ The system is very important, for fixing records.

FR12: Data Access and Retrieval

The system needs to let people who are allowed to use it look up information about people, in the population records. These people should be able to search using different things to find what they are looking for. They can use population records to find the information they need.

The system will help us look at population data in ways. We can use it to see the data for administrative locations. We can also use it to see the data for demographic attributes. The system will support filtering population data by location and demographic attributes.

This means we can look at the population data for an administrative location and also look at the population data, for specific demographic attributes. The population data can be filtered by location and demographic attributes.

The system needs to get population information correctly. The population information has to be retrieved by the system. This is important for the population information to be accurate when the system gets it. The system has to do this for the population information.

FR13: Statistical Analysis Support

The system will give us information about the people in a population. This includes things like how old people're whether they are men or women and how the population is changing over time. The system will show us population statistics, like age distribution it will show us gender distribution. It will also show us population growth trends.

The system will let us put together statistics at administrative levels like at the city level or the country level so we can see the statistics at different administrative levels. This means we can look at the statistics at administrative levels, such as the statistics at the city level or the statistics, at the country level and the system will allow us to do this for all the different administrative levels.

The system needs to help us compare population data from times. We want to look at population data over time and see how it changes. The system should be able to do this for time periods. We are talking about the population data and the system has to support comparison of this population data.

FR14: Data Export and Interoperability Support

The system needs to let people who are allowed to use it get the population information and reports in a format that's easy to understand. This means people who have permission can export the population data and reports in formats, like the ones we normally use.

The system has to make sure that the data it sends out follows the rules for who can see it. The system must check the data it exports to make sure it complies with the access control rules. This means the system has to look at the data it is exporting and make sure that it is allowed to share that data, with the people it is sending it to. The system has to follow these access control rules when it exports data.

The system needs to work with government information systems in the future. This means the system has to be able to share information with government systems. The system should be able to talk to government information systems. The system must support work, with other government information systems like the ones the government uses.

FR15: System Notification and Alerts

The system will let users know when they have verification tasks that are waiting to be done. The system does this to remind users about these pending verification tasks. Users need to complete these verification tasks. The system helps them remember to do so.

The system will tell the administrators when it sees something that does not look right like someone trying to log in times and failing. The system is looking for activities so it will alert the administrators about these suspicious activities. This way the administrators can take a look, at the suspicious activities and the repeated failed login attempts.

The system needs to send reminders that the system makes on its own. These reminders are about when we need to update the data in the system. The system reminders will help us keep the data in the system up, to date.

FR16: Backup and Data Recovery Operations

The system will have to support doing backups of our data on a schedule. This means the system has to be able to do these scheduled data backup operations.

The system needs to let people who're, in charge and have permission like administrators get back their data from the backups.

This means administrators will be able to use the system to restore their data when they need to. The system has to make sure authorized administrators can do this with the backups.

The system needs to make sure that we can get our data back if the system fails. This is important for the system to do because we need to be able to recover our data when something goes wrong with the system. The system has to have a way to get our data back so we do not lose it when the system fails.

3.7.2 Non-Functional Requirements

The Population Count System has some -functional requirements that tell us about the quality of the system and what it can and cannot do. These non-functional requirements of the Population Count System define what makes the Population Count System good and what limits the Population Count System has.

NFR1: Security Requirements

The system must keep the population data secret. It has to make sure that the population data is not shared with anyone who should not see it. The system has to protect the population data at all times. The population data must be kept confidential by the system.

The system has to keep the data safe from people who are not supposed to see it. The system must protect the data at all times. This means the system has to stop people from getting to the data. The system is responsible, for keeping the data safe from access.

The system needs to have ways for people to log in. This means the system has to support authentication mechanisms so that people can be sure their information is protected when they use the system and its secure authentication mechanisms.

The system needs to keep a record of when people use the system and when they do things with the system. The system has to log every time someone gets into the system and every action that happens with the system.

NFR2: Performance Requirements

The system needs to answer the user requests quickly. The user requests should get a response from the system within an amount of time. This is important for the system to work properly and help the user. The system has to make sure it responds to the user requests, in a manner.

The system needs to let many users work on it at the time. This means the system has to support people using it all at once which is really important, for the system. The system will have to handle this so that multiple users can access the system without any problems. The system is supposed to do this.

The system needs to be able to deal with more population data without getting slower. The population data volume is going to keep increasing. The system has to handle it without any performance issues. The system should still work well even when it has to process a lot of population data.

NFR3: Reliability and Availability

The system has to make sure that it stores information about people in a way that's safe and always works. The population data needs to be stored in a way, by the system.

The system should try to keep data safe if the system fails. The system has to make sure that we do not lose a lot of data when the system is not working properly. This is very important, for the system. The system needs to protect our data in case something goes wrong with the system.

The system needs to have a way to back up data and get it back if something goes wrong with the data. This means the system has to be able to support data backup and also data recovery mechanisms. The data backup is important. The system has to do it. The system also has to be able to recover the data when it is needed.

NFR4: Usability Requirements

The system needs to be easy to use for everyone. This means the system has to be simple enough for people who're not good with computers.

The system should be easy to understand and use, for people who do not know a lot about technology. The system is for users, with levels of technical skills so the system has to work well for all of them.

The system is going to use labels that're easy to understand and workflows that make sense. The system will have labels so that users know what they are doing. The system will also have structured workflows to help people get things done.

The system will make it easier for people to register and update their information. It will do this by reducing the amount of work that needs to be done by hand during population registration. When people need to update their details. The system is designed to simplify the process of population registration and updates.

NFR5: Scalability

The system needs to be able to support expansion so it can accommodate the growth of the population. The population is going to keep growing. The system has to be able to grow with the population. This means the system has to support expansion.

The system will be able to support administrative units when we need them. The system needs to be able to handle administrative units. This is important, for the system and the administrative units.

The system will have to work with the systems that we will have in the future. This means the system must be able to integrate with these national systems. The national systems that are coming will need to be able to work with the system.

NFR6: Maintainability

The system needs to be simple so it is easy for people to take care of it and make changes to it. The system should be easy to update. This way the system will always be easy to maintain. The system will not be a problem.

The system needs to let us make changes, to the system features without causing problems.

We should be able to modify the system features and the system will still work properly. This means the system features can be updated and the system will keep running.

The system needs to let people who're in charge and have permission to make changes, to the system configurations. This means that authorized administrators can do things like manage the system configurations. The system configurations are what the system uses to work and the administrators will be able to make changes to these configurations.

NFR7: Legal and Ethical Compliance

The system has to follow the rules of the country when it comes to keeping peoples information safe and private. The system must do what the national data protection and privacy policies say. This means the system has to be careful, with data protection and privacy policies.

The system must make sure it handles information about people in a fair and honest way. The system has to deal with population data in a very careful manner. The system is responsible, for taking care of population data.

3.7.3 System Constraints

The system will work with the computer systems that're already in place, in Ethiopia. The system is going to use the internet and technology that Ethiopia already has. This means the system will be a part of Ethiopias existing computer infrastructure. The system will operate within this setup.

The system is for use by government institutions that have permission. These government institutions must be authorized to use the system. Only government institutions, with the permission can use the system.

The system shall rely on software-based solutions without specialized hardware requirements.

3.8 Assumptions and Constraints

This part talks about the things we assumed and the limits we considered when we were figuring out what the Population Count System needs to do. These things tell us how the Population Count System is supposed to work and what the Population Count System can and cannot do. They also affect what the Population Count System needs. If it is possible to make it work.

3.8.1 Assumptions

- The government is going to help with this project. Government institutions, at the level the regional level and the local level will work together to give us the population data we need and support the system. The government institutions will provide the help to make the system work properly.
- People who are allowed to use the system will get some training on how to use the system before they actually start using it. They will learn the basics of the system so they know what they are doing when the system is deployed. The system is pretty important so the people, in charge want to make sure that the users know what they are doing with the system.
- People think that the offices where administrators work will have the things they need like computers and a way to get on the internet. The administrative offices will have Information and Communication Technology infrastructure, such, as computers and network access that the staff can use.
- People think that the information, about how many people live in a place that is put into the system will match the lines that separate different areas. The population data will follow the boundaries that the government uses. The population data is what is collected and then entered into the system.
- People think that the countrys rules about keeping information private and managing the population will help with using a big computer system that stores all the information in one place. This computer system is like a database that holds all the information about the population and it is controlled from one central location. The rules, about data privacy and population management are very important for this system to work properly. The countrys rules will support the use of this centralized system.

- The system is mostly going to be used by government institutions. This means that the general public will not be using the system directly. The government institutions will be the users of the system.

3.8.2 Constraints

- The system has to work with what Ethiopia has, in terms of information and communication technology and the way things are managed. The system is limited to Ethiopias existing information and communication technology infrastructure and the way the country is administered.
- The system only works with software solutions. It does not work with things or special hardware. The system is limited to software. It does not include things, like specialized hardware integration.
- We have to work with a budget. This means we cannot use technologies or special software that costs a lot of money. The budget limitations are a problem when it comes to using these advanced or proprietary technologies.
- The system has to follow the rules, about keeping peoples information safe and private in our country. The system must do this to meet data protection and national confidentiality requirements.
- There are some places where the internet does not work well. This can be a problem when you want to use the system away. The internet connectivity limitations, in some regions may affect real-time system access. So you might not be able to get into the system when you need to because of the internet connectivity limitations.
- The project timeline is constrained by academic and institutional schedules.

3.10 Requirements Prioritization

The Population Count System has a lot of requirements that need to be met. So we prioritize them to make sure the important ones are taken care of first. We have to consider the project and the challenges that Ethiopia faces with administration and technology. This helps us balance what the system needs to do what is possible to do and what the people involved really need from the Population Count System.

The Population Count System has a lot of needs that're very important. These needs are ordered by how important they're, to the system working properly.

We also think about how they affect the way the country manages its population.. We look at how these needs are connected to each other. The Population Count System requirements are a deal because they help the Population Count System do its job.

3.10.1 High-Priority Requirements

The Population Count System needs some things to work properly. These things are very important for the system to do its job. The Population Count System will not work well without these things. The Population Count System requires these things to be, in place so it can function the way it should. The Population Count System is counting on these things to operate correctly.

- ✓ User authentication and role-based access control
- ✓ Population registration and household management
- ✓ Population data update (birth, death, migration)
- ✓ Centralized population data storage
- ✓ Data validation and duplicate detection
- ✓ Population data verification and approval
- ✓ Basic population reporting by administrative level
- ✓ Security and confidentiality of population data

The things that are needed for this project are very important for making sure we get the number of people, in Ethiopia. This is necessary for the Ethiopian government to accept and use the system. The Ethiopian government needs these requirements to be able to use the system.

3.10.2 Medium-Priority Requirements

The medium-priority requirements make the system work better and are easier to use. You do not need them for the system to work when you first start it. The medium-priority requirements are important for the system and they help with system efficiency and usability. The system is more efficient and easier to use when you have the medium-priority requirements, in place.

- ✓ Advanced population statistics and trend analysis
- ✓ Audit trail and activity logging

When you want to share information you need to export the data. Then share the report. Data export is a step because it helps you to get the data out of the system. After that you can share the report, with others. Data export and report sharing are useful tools.

You can use data export to get the data. Then use report sharing to send it to someone else. Data export and report sharing make it easy to work with others and share information.

- ✓ System notifications and alerts
- ✓ Administrative hierarchy configuration
- ✓ Backup and data recovery support

These requirements make the system management better people are more accountable. The system can analyze things more easily. The system management is. The analytical capabilities of the system are also better.

3.10.3 Low-Priority Requirements

Low priority requirements are things that we can add to the system on. We do not need these priority requirements for the system to work properly. The system will still work fine without these priority requirements. We can add these priority requirements in later phases. This way the core system functionality will not be affected by the priority requirements. The low priority requirements are not that important for the system to function.

- ✓ Advanced analytical dashboards
- ✓ Integration with future national systems
- ✓ Extended reporting customization
- ✓ Advanced system configuration options

These requirements help the system grow over time and make it easier to add things to the system in the future. The system will be able to expand. The requirements will support these future enhancements to the system.

3.10.4 Prioritization Rationale

The prioritization approach ensures that the Population Count System initially focuses on core population data management functions, followed by system optimization and expansion features.

This approach aligns with Ethiopia's current operational environment and supports phased system development while maintaining system reliability and usability.

3.11 Requirements Traceability

Requirements traceability ensures that every requirement of the Population Count System is linked to its source, stakeholder, and corresponding system function. This enables verification, validation, and management of changes throughout the system lifecycle.

3.11.1 Traceability Objectives

- Ensure all requirements are derived from stakeholder needs, business processes, and system objectives.
- Facilitate verification that each requirement is implemented correctly in the system.
- Provide a clear mapping between functional and non-functional requirements, stakeholders, and business processes.
- Support impact analysis when requirements change or are updated.

3.11.2 Traceability Matrix (Summary Example)

Requirement ID	Requirement Description	Source / Stakeholder	Functional Area	Priority
FR1	User Authentication and Authorization	Federal & Regional Administrations	Security / Access	*High*
FR2	Population Registration	Local Registration Officers	Data Entry	*High*
FR3	Population Data Update	Local & Regional Offices	Data Maintenance	*High*
FR4	Population Data Verification	Supervisory Users	Verification	*High*
FR5	Population Data Management	Federal Administration	Database Management	*High*
FR6	Reporting and Data Aggregation	Planners / Policy Makers	Reporting	*High*
FR7	Administrative Management	System Administrators	Administration	*Medium*
FR8	Administrative Hierarchy Management	Federal & Regional Administrations	Data Structure	*High*
FR9	Data Validation and Integrity Control	Registration Officers	Data Quality	*High*
FR10	Audit Trail and Activity Logging	Federal Administration	Security / Compliance	*High*
FR11	Duplicate Detection and Record Matching	Registration Officers	Data Quality	*Medium*
FR12	Data Access and Retrieval	Local & Regional Offices	Search / Retrieval	*High*
FR13	Statistical Analysis Support	Planners / Policy Makers	Reporting / Analytics	*Medium*
FR14	Data Export and Interoperability Support	Federal Administration	Reporting / Integration	*Medium*
FR15	System Notification and	System	Operations / Alerts	*Medium*

	Alerts	Administrators		
FR16	Backup and Data Recovery Operations	System Administrators	Data Management	*High*
FR17	User Support and Help Functions	Registration Officers	Usability	*Medium*
FR18	System Configuration Management	System Administrators	Administration	*Medium*
NFR1	Data Security & Privacy	All Stakeholders	Non-functional	*High*
NFR2	Performance	All Stakeholders	Non-functional	*High*
NFR3	Reliability & Availability	All Stakeholders	Non-functional	*High*
NFR4	Usability	Registration Officers / Planners	Non-functional	*Medium*
NFR5	Scalability	Federal Administration	Non-functional	*Medium*
NFR6	Maintainability	System Administrators	Non-functional	*Medium*
NFR7	Legal & Ethical Compliance	Federal Administration	Non-functional	*High*

Notes:

- ✧ "Priority" is classified as High, Medium, or Low based on stakeholder impact and system criticality.
- ✧ This matrix provides a foundation for traceability from requirements → stakeholders → business processes → validation.

3.11.3 Benefits of Traceability for the Project

- Enables verification that all population management needs in Ethiopia are captured.
- Supports requirement validation before system implementation.

- Facilitates impact assessment if population policies, administrative structures, or data processes change.
- Improves system maintainability by linking requirements to stakeholders and business objectives.

CHAPTER FOUR

SYSTEM MODELING AND DESIGN

4.1 System Architecture Overview

The population count system has a plan that shows how all the parts work together and how information moves around. This plan is important because it helps the population count system grow be easy to maintain and work correctly when it collects, stores and processes population data. The population count system needs to be able to handle people and information over time so this plan is very useful.

The system is made up of four layers. At the top we have the users, which include Admins and Enumerators and Citizens. These users work with the system using a website.

The part that people see which is called the presentation layer or the UI layer has forms that the users can fill out to put in information. It also has dashboards that show what is going on and tools that help with reporting and looking at the information, which's like analytics, for the system and the users, including Admins and Enumerators and Citizens.

The application layer is where the business logic of the system happens. It does things like authentication and data validation. It also does aggregation and reporting. This layer takes care of all the things that users put into the system. It checks to make sure everything is correct. Then gets the information ready to be stored or shown to people.

The database layer is like a storage box for the system. It keeps all the information, about users and census forms and households and the people who live in those households. The database makes sure that all the information is connected correctly and that it is accurate. It also helps get the information when it is needed.

The interaction between the layers is really smooth. Users put information into the system using the user interface. This information goes to the application layer where it gets worked on. The application layer then talks to the database to either store information or get some back.

Then the application layer sends responses like reports or messages saying everything is okay back through the user interface, to the users. This makes sure that the data stays consistent and correct all the way through the process. The application layer and the database and the user interface all work together to make this happen with the application layer and the database and the user interface.

This system is set up in a way that makes it easy to understand and work on. It is designed so that the system can handle users or areas without any problems. The system is very reliable because it checks everything carefully. It is also very secure because it has rules about who can do what and it controls who can get in.

The system is divided into parts like the interface, the logic and the data. This makes it easier to fix things and add things. At the time the system is quick and secure for all users of the system. The system is easy to take care of because it is divided into these parts. This makes it a good system, for all users of the system.

The professional diagram generated earlier visually represents this architecture and can be included directly in the project report:

The **component interaction table** summarizes how each part works together:

Component	Input	Output	Interaction
Users	Actions, form input	Requests for processing	Sends input to UI
UI Layer	User input	Form data, display info	Interacts with Logic Layer
Application Layer	Validated input	Processed data, reports	Interacts with Database
Database Layer	Queries	Stored/retrieved data	Provides data to Logic Layer

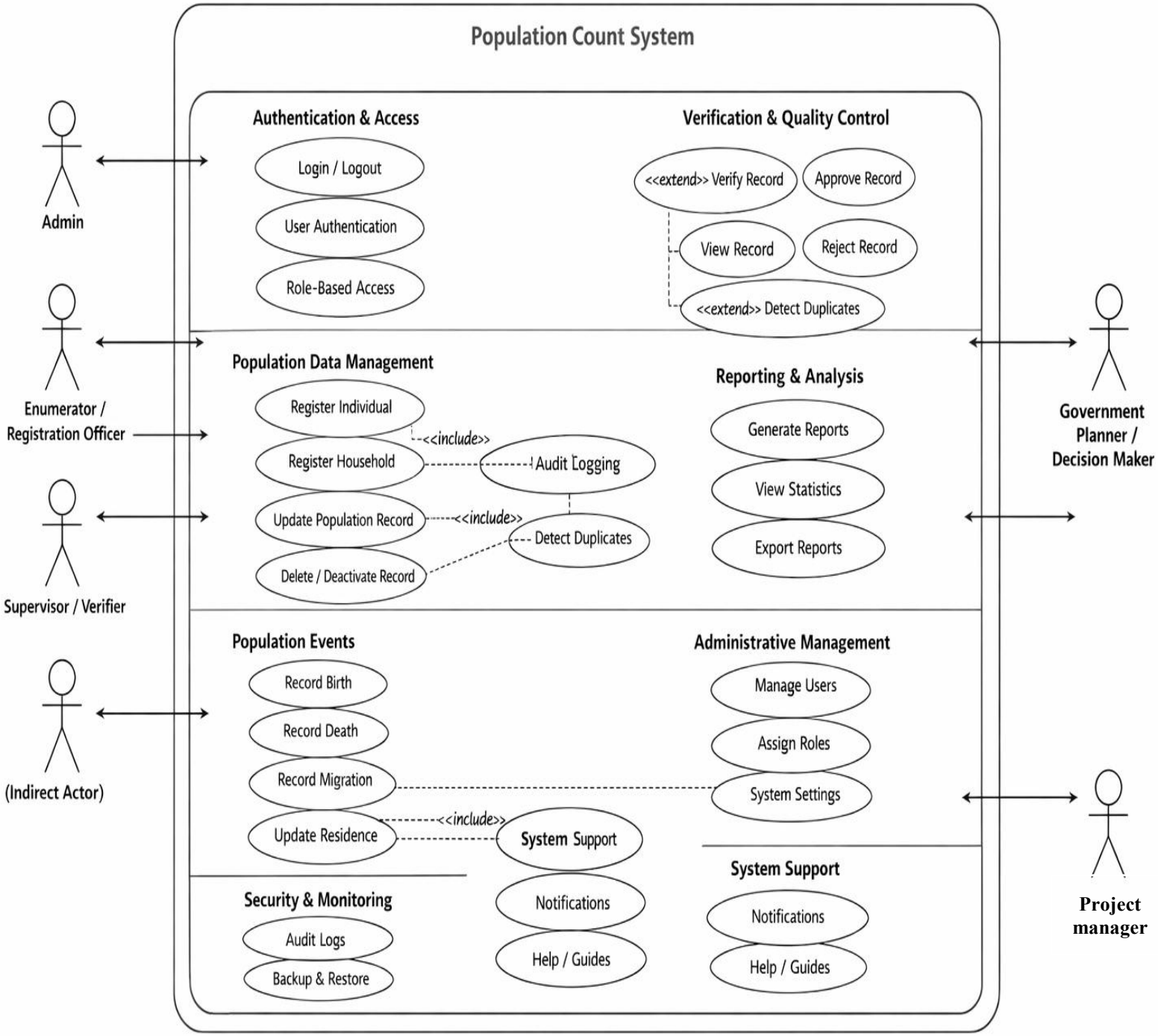
4.2 Use Case Diagram and Descriptions – Population Count System

Actors :-

- ✓ Admin: Manages users, system configuration, generates reports, oversees operations.
- ✓ Enumerator: Collects household/citizen data, submits census forms.
- ✓ Citizen: Provides personal information, can verify/update their data (if allowed).
- ✓ Supervisor/Verifier: Checks, approves, or rejects submitted population records.

Use Cases :-

- ✓ Login / Logout
- ✓ Submit Census Form
- ✓ Register Population Data
- ✓ Update Population Data
- ✓ Verify Population Records
- ✓ Generate Reports
- ✓ Manage Users and Roles
- ✓ Manage Administrative Hierarchy
- ✓ Audit Trail / Activity Logging
- ✓ Duplicate Detection



4.3 Activity Diagrams

Activity diagrams illustrate the workflow of activities within the Population Count System. They show how actions are performed step by step, including decision points, parallel processes, and the sequence of operations from start to end.

These diagrams help visualize business processes and system behavior during population data collection, verification, and reporting.

Activity Diagram: User Login Process

Description:

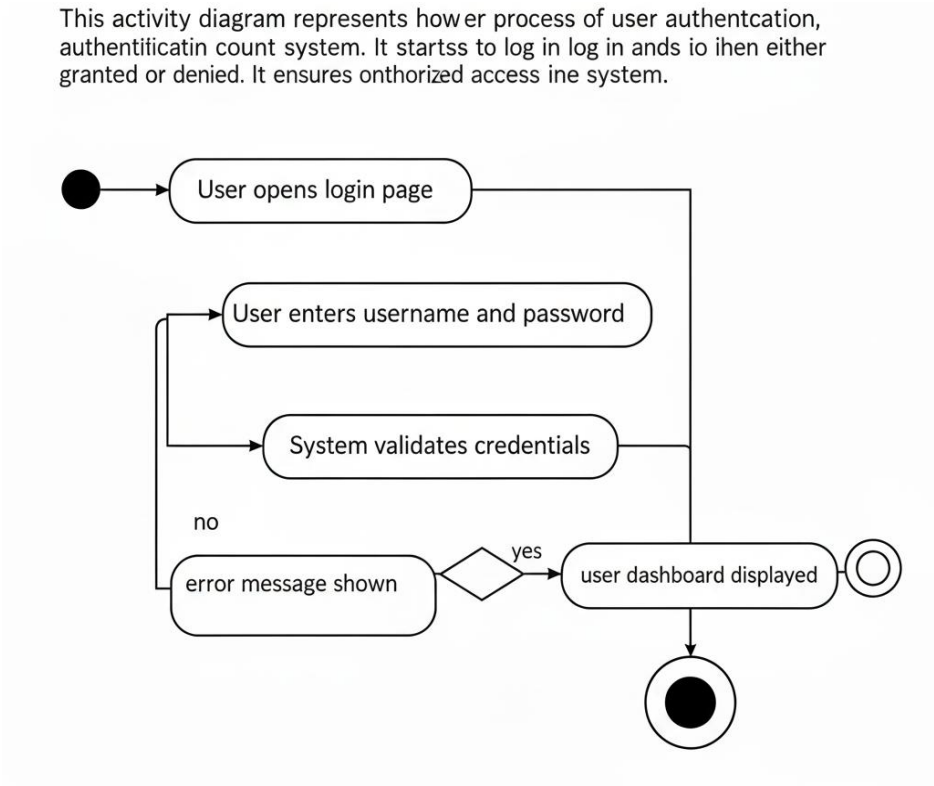
This activity diagram represents the process of user authentication. It starts when the user attempts to log in and ends when access is either granted or denied.

Activity Flow:

- ✓ User opens login page
- ✓ User enters username and password
- ✓ System validates credentials
- ✓ Decision: credentials valid?
  - If yes → user dashboard displayed
  - If no → error message shown

Purpose:

- ✓ Ensures only authorized users access the system.



Activity Diagram: Household and Population Data Registration

Description:

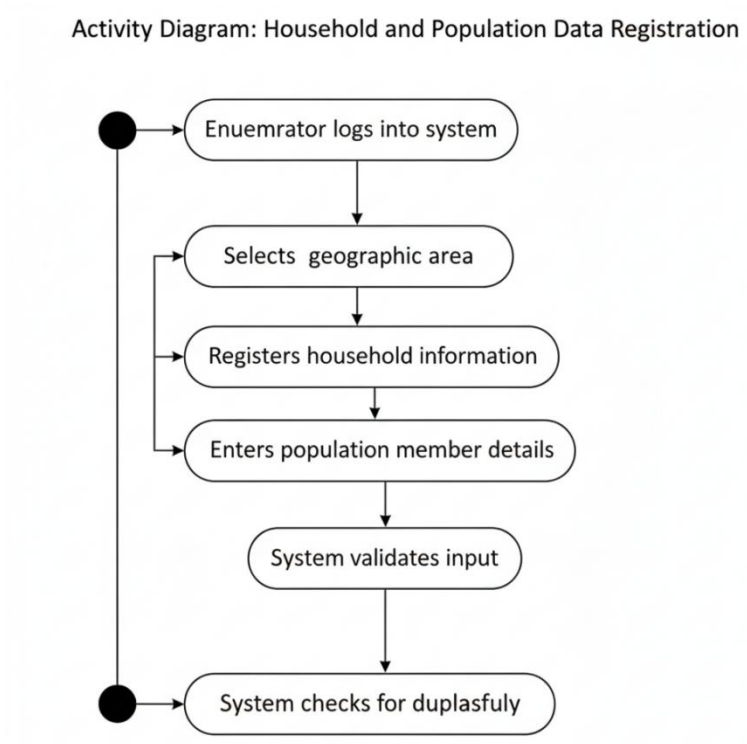
This diagram shows how an enumerator registers a household and its members during census data collection.

Activity Flow:

- ✓ Enumerator logs into system
- ✓ Selects geographic area
- ✓ Registers household information
- ✓ Enters population member details
- ✓ System validates input
- ✓ System checks for duplicates
- ✓ Data stored successfully

Purpose:

Represents accurate and structured population data entry.



Activity Diagram: Census Form Submission

Description:

This activity diagram illustrates the process of submitting completed census data for verification.

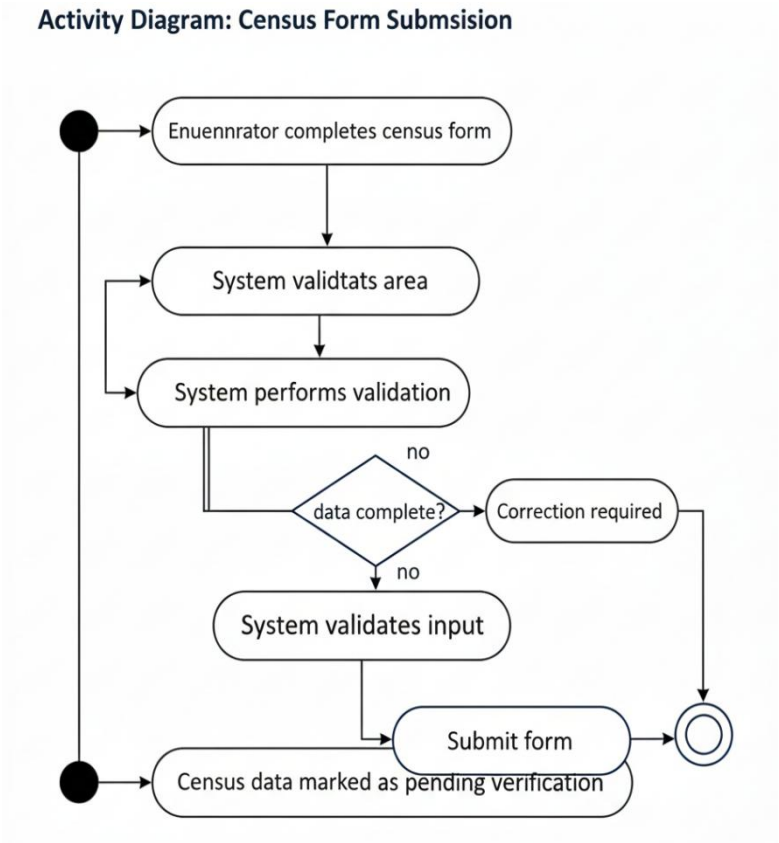
Activity Flow:

- ✓ Enumerator completes census form
- ✓ System performs validation
- ✓ Decision: data complete?
    - If no → correction required
    - If yes → submit form

Census data marked as pending verification

Purpose:

- ✓ Ensures only complete and valid census data is submitted.



Activity Diagram: Population Data Verification

Description:

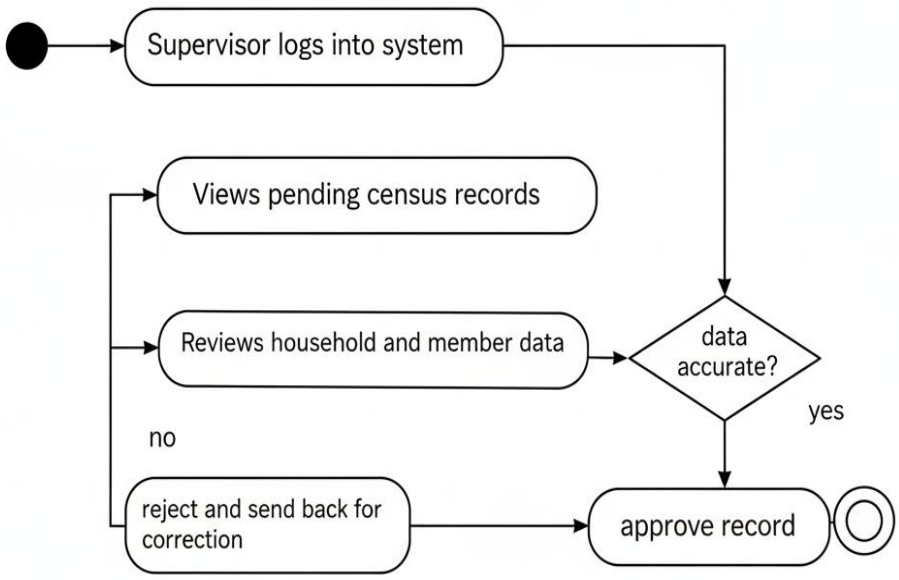
This diagram describes how supervisors verify submitted population records.

Activity Flow:

- ✓ Supervisor logs into system
- ✓ Views pending census records
- ✓ Reviews household and member data
  - Decision: data accurate?
  - If yes → approve record
- ✓ If no → reject and send back for correction

Purpose:

- ✓ Maintains data quality and integrity.



Activity Diagram: Population Data Update

Description:

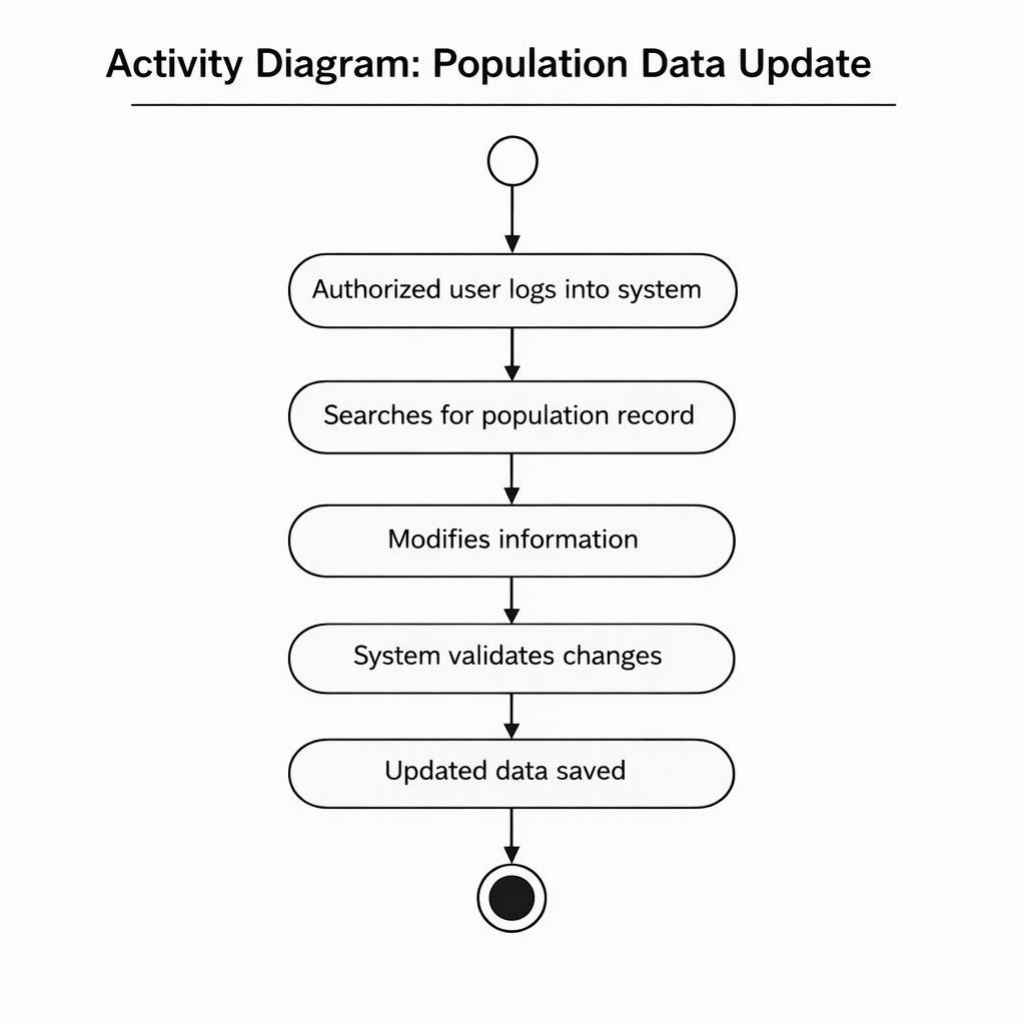
This diagram shows the process of updating existing population records.

Activity Flow:

- ✓ Authorized user logs in
- ✓ Searches for population record
- ✓ Modifies information
- ✓ System validates changes
- ✓ Updated data saved

Purpose:

Allows correction and updating of population information over time.



Activity Diagram: Report Generation

Description:

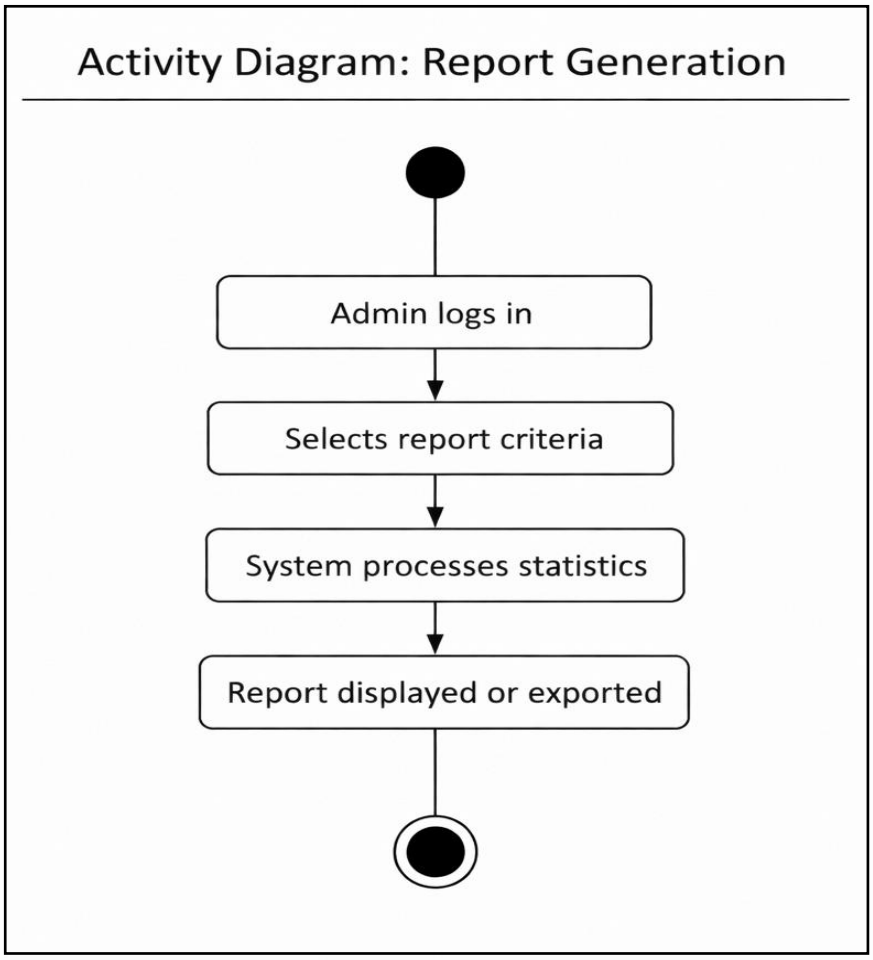
This activity diagram represents the process of generating population statistics and reports.

Activity Flow:

- ✓ Admin logs in
- ✓ Selects report criteria
- ✓ System retrieves data
- ✓ System processes statistics
- ✓ Report displayed or exported

Purpose:

Supports decision-making and planning through data analysis.



Activity Diagram: User Management

Description:

This diagram shows how administrators manage system users and roles.

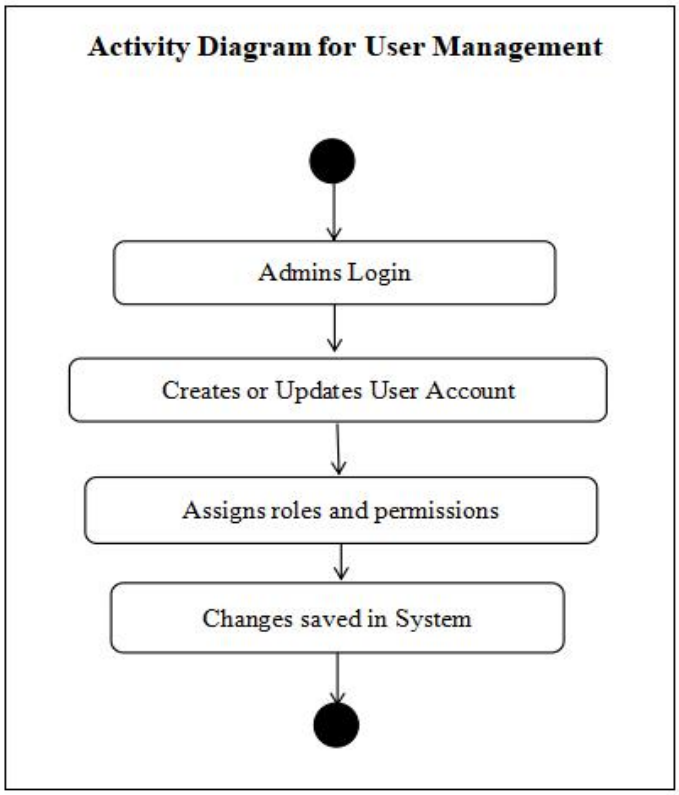
Activity Flow:

- ✓ Admin logs in
- ✓ Creates or updates user accounts
- ✓ Assigns roles and permissions
- ✓ Changes saved in system

Purpose:

- ✓ Controls system access and security.
- ✓ Why Activity Diagrams Matter in Your Project
- ✓ They clearly show workflow logic
- ✓ They match use cases and system functions

- ✓ They make your design easy to understand for evaluators
- ✓ They help later when implementing code
- ✓ Detailed Use Case Descriptions



Use Case	Actor	Precondition	Flow of Events	Postcondition
Login	All Users	Must be registered	Enter credentials → System validates → Access granted	User logged in
Register Population Data	Enumerator	Logged in	Enter household data → Validate → Submit	Data stored in database
Submit Census Form	Enumerator	Logged in	Fill form → Validate → Submit	Census form saved
Update Population Data	Enumerator / Admin	Logged in	Locate record → Modify data → Validate → Save	Updated record saved
Verify Population Records	Supervisor	Logged in	Review submitted records → Approve/Reject	Verified records updated
Generate Reports	Admin	Logged in	Select report type → System processes → Display / Export	Report generated
Manage Users / Roles	Admin	Logged in	Create/modify/deactivate users → Assign roles	User access updated
Audit Trail / Activity Logging	Admin	Logged in	Access audit log → Review actions	Activity log viewed
Duplicate Detection	Enumerator / Admin	Logged in	System checks for duplicates → Notify user	Duplicate records flagged

4.4 Sequence Diagrams

Sequence diagrams show us how the different parts of a system talk to each other. They do this in the order that it happens.

Sequence diagrams show how a user sends a request and then what happens to that request. The system gets the request does something with it. Then sends a response back.

This helps us see how the user interface, the application logic and the database all work together when the system is running.

Sequence diagrams are really good, at highlighting how messages move between the user interface the application logic and the database.

Sequence Diagram: User Login

Participants:

User → User Interface → Authentication Service → Database

Interaction Flow:

The user enters their login credentials to get into the system. They type in their username and password to access the account. The user is trying to log in with their login credentials.

The user interface sends the credentials to the authentication service. This is what happens when the user interface is sending the credentials to the authentication service. The user interface and the authentication service work together and the user interface sends the credentials to the authentication service so it can check if everything is okay.

The authentication service looks at the database to get the information it needs. It checks the database to make sure everything is okay. The authentication service is trying to verify some details so it queries the database.

The database gives us the result of the validation. This result is what the database returns after it checks everything. The database returns this validation result to let us know if everything is okay.

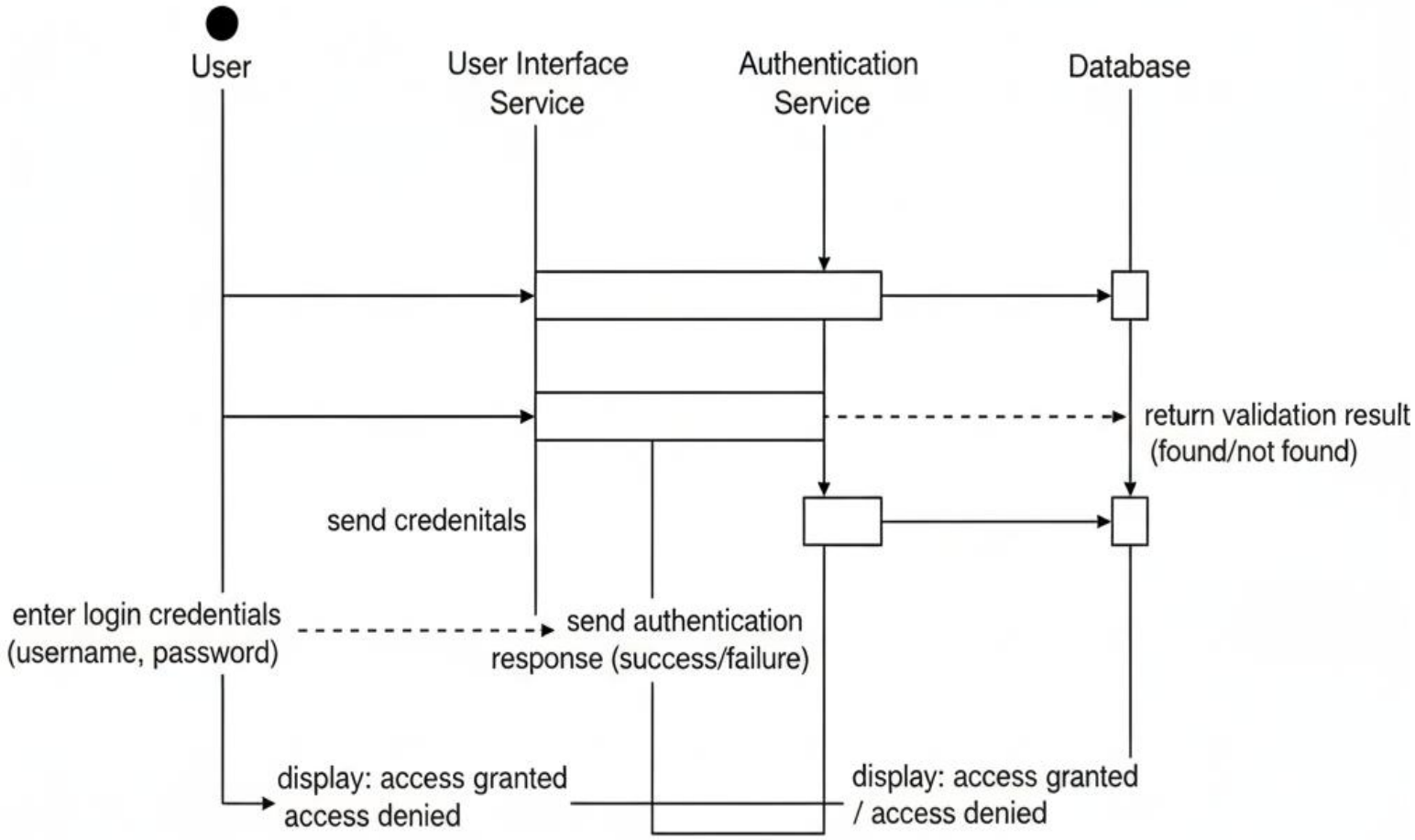
The authentication service sends a response to the user interface. This is how the authentication service and the user interface work together. The authentication service is very important because it helps the user interface to know if someone is allowed to use the system. The authentication service does this by checking the users information. The user interface then gets a response, from the authentication service. The authentication service is used to make sure that only the right people can use the system. The authentication service responds to the user interface so that it can let the user in or keep them out.

UI grants or denies access

Purpose:

✓ Ensures secure authentication and role-based access.

Sequence diagram for user login



Sequence Diagram: Register Household and Population Data

Participants:

Enumerator → User Interface → Data Validation Service → Database

Interaction Flow:

The enumerator goes into a house. Writes down information about the people who live there. They ask questions. Enter details about the household. The enumerator is doing this to get information, about the household.

The user interface forwards the data to the validation service. This is what the user interface does with the data it forwards it to the validation service for checking. The validation service then looks at the data, from the user interface.

The validation service is used to check if the data's consistent. It makes sure the data consistency is okay. The validation service does this by checking the data. This is what the validation service does for data consistency.

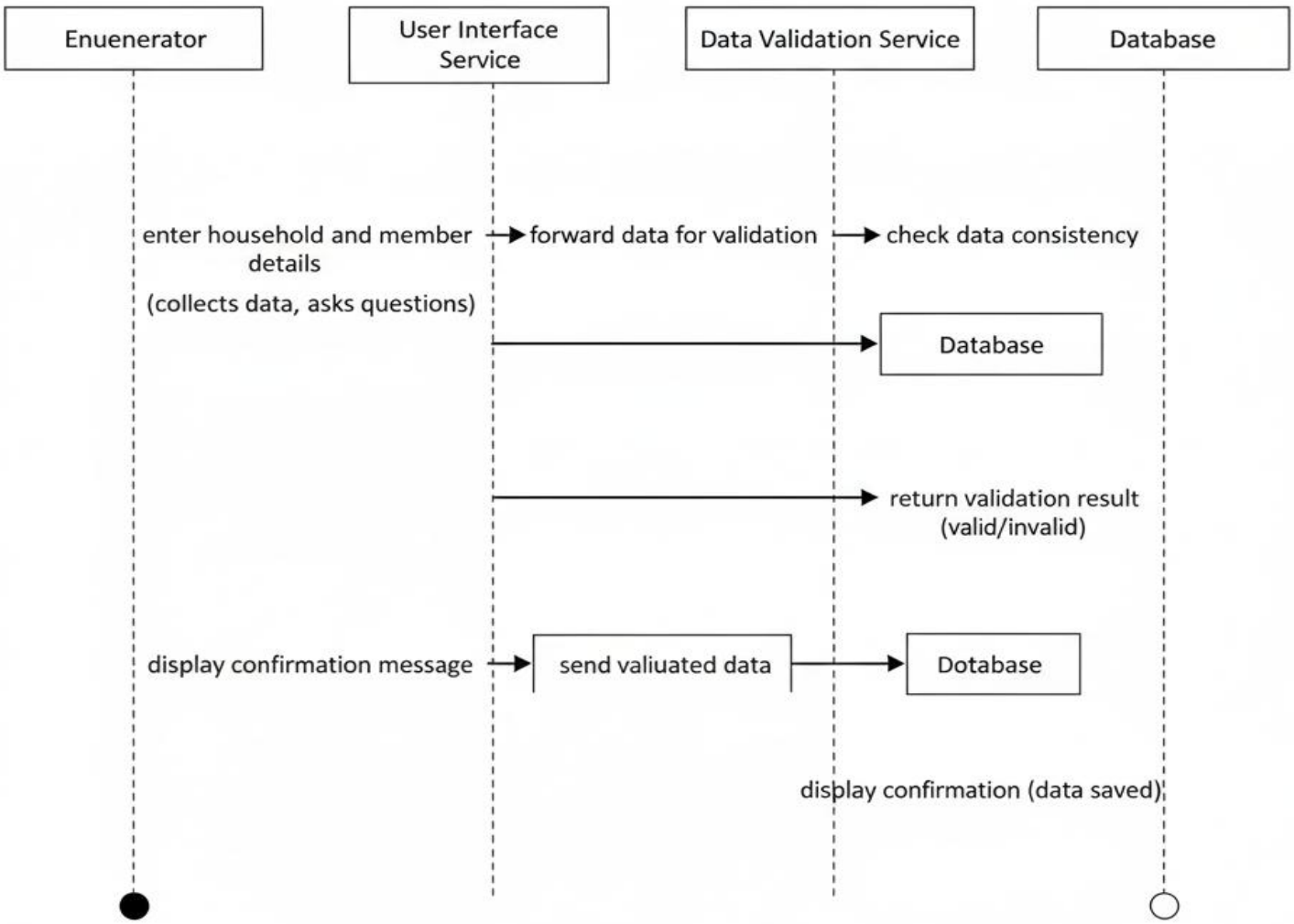
Your information is now stored in the database. The database has your data saved.

The enumerator got a confirmation back. This confirmation was sent to the enumerator. The enumerator received the confirmation.

Purpose:

- ✓ Ensures accurate population data entry.

Sequence Diagram : register Household and Population



Sequence Diagram: Submit Census Form

Participants:

Enumerator → User Interface → Census Processing Service → Database

Interaction Flow:

The enumerator is done with the census form. Now they submit it. They have finished filling out the census form. It is complete. The enumerator submits the completed census form.

The user interface sends the form data so it can be processed. The form data that the user interface sends is used for processing.

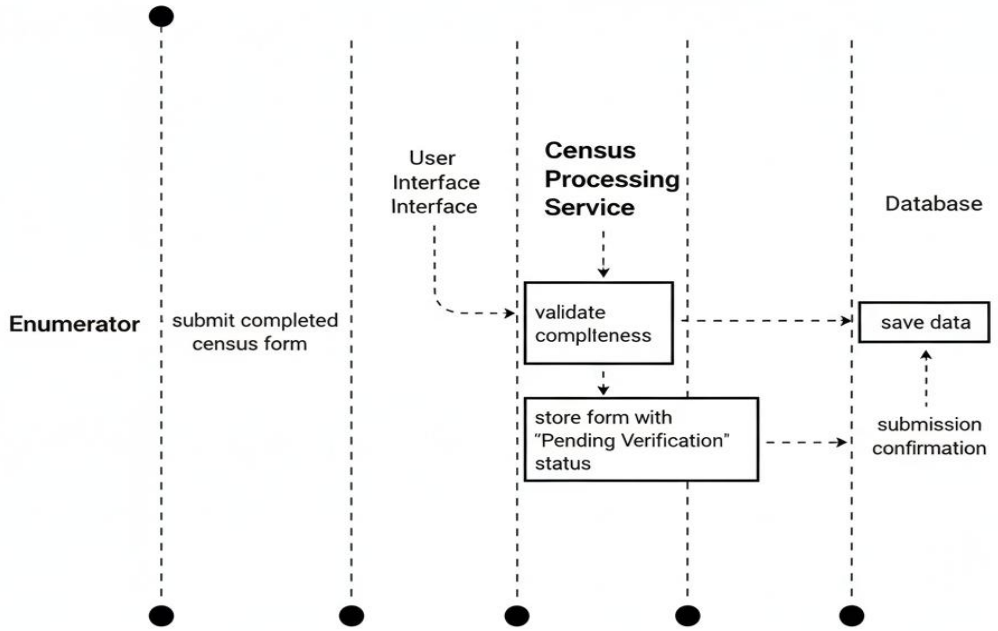
The system checks to make sure everything is complete. It looks at all the information to see if the system has everything it needs. The system validates completeness of all the details.

Census form stored with “Pending Verification” status

The system says that the submission confirmation has been returned. This means that the submission confirmation is now back. The submission confirmation. It is here now.

Purpose:

- ✓ Prepares census data for verification.



Sequence Diagram: Verify Population Records

Participants:

Supervisor → User Interface → Verification Service → Database

Interaction Flow:

Supervisor requests pending records

The system gets information from the database. It goes into the database to find the data it needs. The database is where all the information is stored and the system retrieves data, from this database.

Supervisor approves or rejects record

The database has been updated with the verification status of the database. This means that the verification status is now part of the database. The database now contains the verification status.

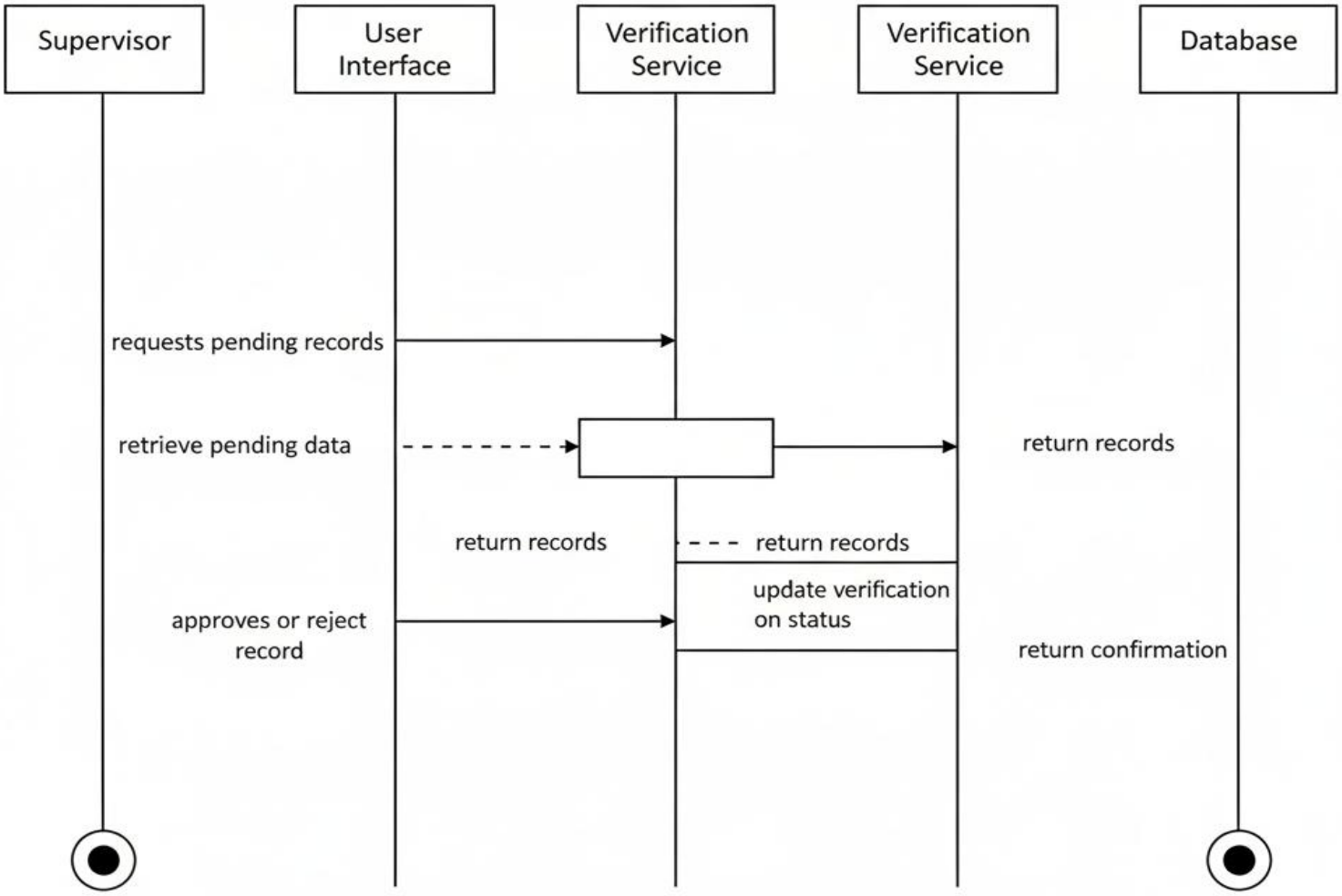
The system is going to send you a confirmation. This confirmation is from the system. You will get this confirmation, from the system.

Purpose:

- ✓ Maintains data accuracy and integrity.

Sequence Diagram : Verify population Records

This diagram shows how supervisors review review and update population records to maintain data accuracy and integrity.



Sequence Diagram: Update Population Data

Participants:

Authorized User → User Interface → Update Service → Database

Interaction Flow:

The user chooses a record that they want to update. They pick the record to make some changes to it. The record that the user selects is the one that they want to modify.

The user interface sends the updated data to the system. This means the user interface is providing information that was not there before. The user interface sends this updated data so that everything is current and accurate. The user interface sends updated data.

The system checks the changes that are made to make sure they are correct. The system validates these changes.

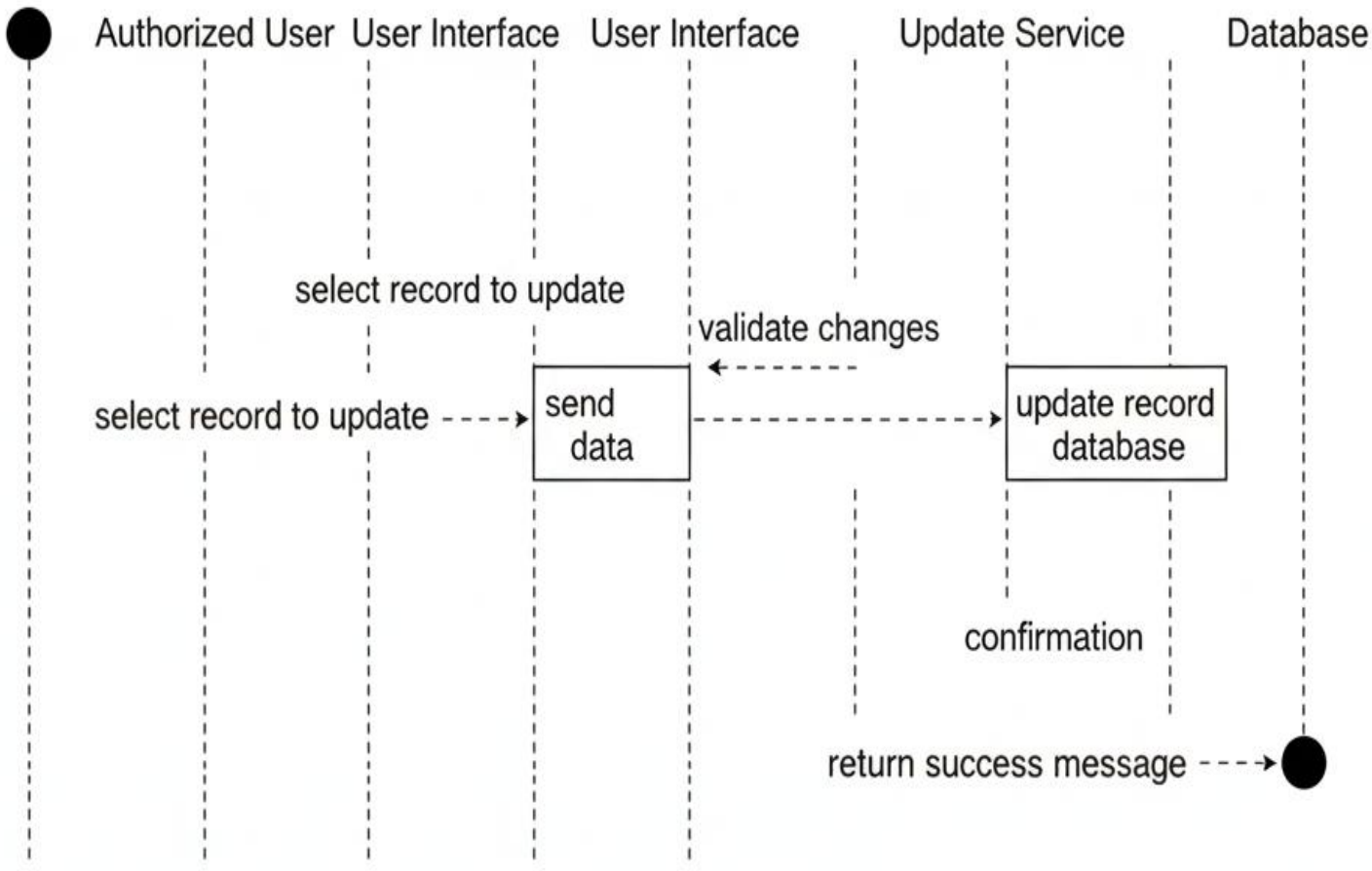
The database record has been updated.

The system says that everything worked out fine and a success message was returned. This means that the thing you were trying to do was completed without any problems and the computer gave you a message to let you know that it was successful. The success message returned is a sign that the success message was received and that the computer is working properly with the success message returned.

Purpose:

This system lets you make changes, to the population records in a controlled way. You can modify the population records.

Sequence Diagram : Update population Data



Sequence Diagram: Generate Population Reports

Participants:

Admin → User Interface → Reporting Service → Database

Interaction Flow:

- ✓ Admin selects report criteria
- ✓ Reporting service requests data

The database gives us the population data. This is the information we get from the database, about the population. We use the database to get the population data.

The system has a lot of statistics about how it works. These statistics are, about the processes that the system uses. The system processes statistics to understand what is going on.

- System processes are very important
- The system uses these statistics to make things better

The system processes statistics every day to see what is working. What is not working. The system processes statistics to help people who use the system.

When you look at the report it is. You can export the report. The report is something that you can see on the screen. You can export the report to use it somewhere else.

Purpose:

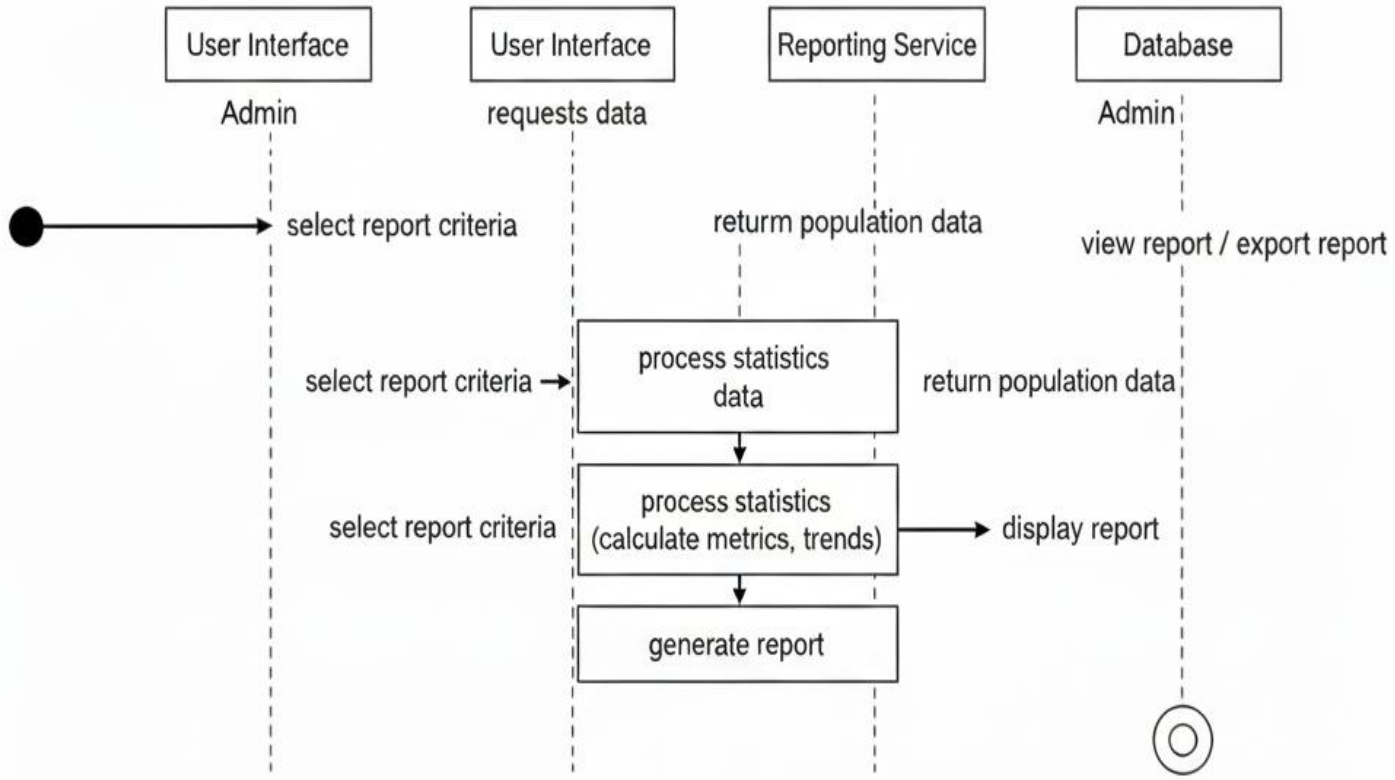
- Supports analysis and decision-making.
- This is a plan to manage people who use the system and the jobs they do.
- The system has to keep track of all the users and what they are allowed to do.
- We make a list of all the users. What roles they have.

  - ✓ We add users to the list when they join the system.
  - ✓ We also change the roles of users when they get a job.

- ✧ The system has to make sure that each user can only do the things that their role allows them to do.
- ✧ We have to be careful and make sure that users do not get access, to things they should not be able to do.
- ✧ The goal is to keep the system safe and make sure that all the users and roles are managed correctly.

Sequence Diagram : Generate Population Reports

This diagram shows how the system supports analysis and decision-making by generating population repeed on selected on selected criteri and role-based access.



Sequence Diagram : Manage users and roles

Participants:

Admin → User Interface → User Management Service → Database

Interaction Flow:

The admin is the person who creates an user account or updates an existing user account. They do this to make sure the user account has the information.

The admin : Updates the user account so that the user can use it to log in. The admin is, in charge of making sure the user account is working properly. The admin. Updates user accounts every day.

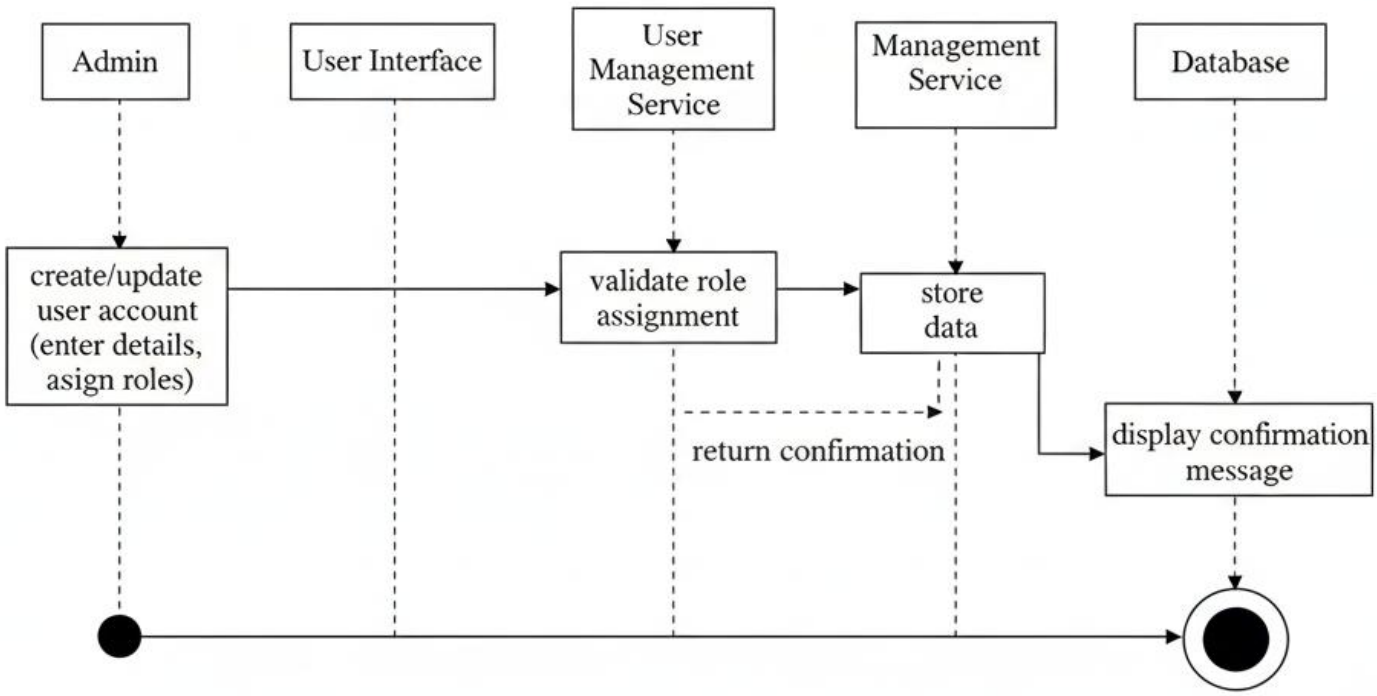
The system checks to make sure the role is assigned correctly to the person. It does this by validating the role assignment. This validation is important for the system to know that the role assignment is done properly. The system validates the role assignment every time.

- ✓ User data stored in database
- ✓ The confirmation has been returned

Purpose:

- ✓ Controls system access and security.

Sequence Diagram : Manage Users And Roles



4.5 Class Diagram

The class diagram shows the Population Count System in a way. It tells us about the classes in the Population Count System and what is part of them. The Population Count System has classes with attributes and methods that do things. The class diagram also shows how these classes, in the Population Count System are connected to each other. This helps us understand how the Population Count System organizes its data and how the parts of the Population Count System work together.

Core Classes and Descriptions

User

Represents all system users such as Admin, Enumerator, Supervisor, and Citizen.

Attributes:

- userID
- fullName
- username
- password
- role
- Status

Operations:

- login()
- logout()
- updateProfile()

Enumerator

Specialized user responsible for collecting census data.

Attributes:

- enumeratorID
- assignedArea

Operations:

- registerHousehold()
- registerMember()
- submitCensusForm()

Supervisor

User responsible for verifying census records.

Attributes:

- supervisorID

Operations:

- verifyRecord()
- rejectRecord()

Admin

User responsible for system administration.

Attributes:

- adminID

Operations:

- manageUsers()
- generateReports()
- viewAuditLog()

Household

Represents a household unit in the census.

Attributes:

- householdID
- address
- region
- registrationDate

Operations:

- addMember()
- updateHousehold()

PopulationMember

Represents an individual person in a household.

Attributes:

- memberID
- fullName
- age
- gender
- relationship
- identificationNumber

Operations:

- updateMember()

CensusForm

Represents a submitted census record.

Attributes:

- formID
- submissionDate
- status

Operations:

- submitForm()
- updateStatus()

Report

Represents population reports generated by the system.

Attributes:

- reportID
- reportType
- generatedDate

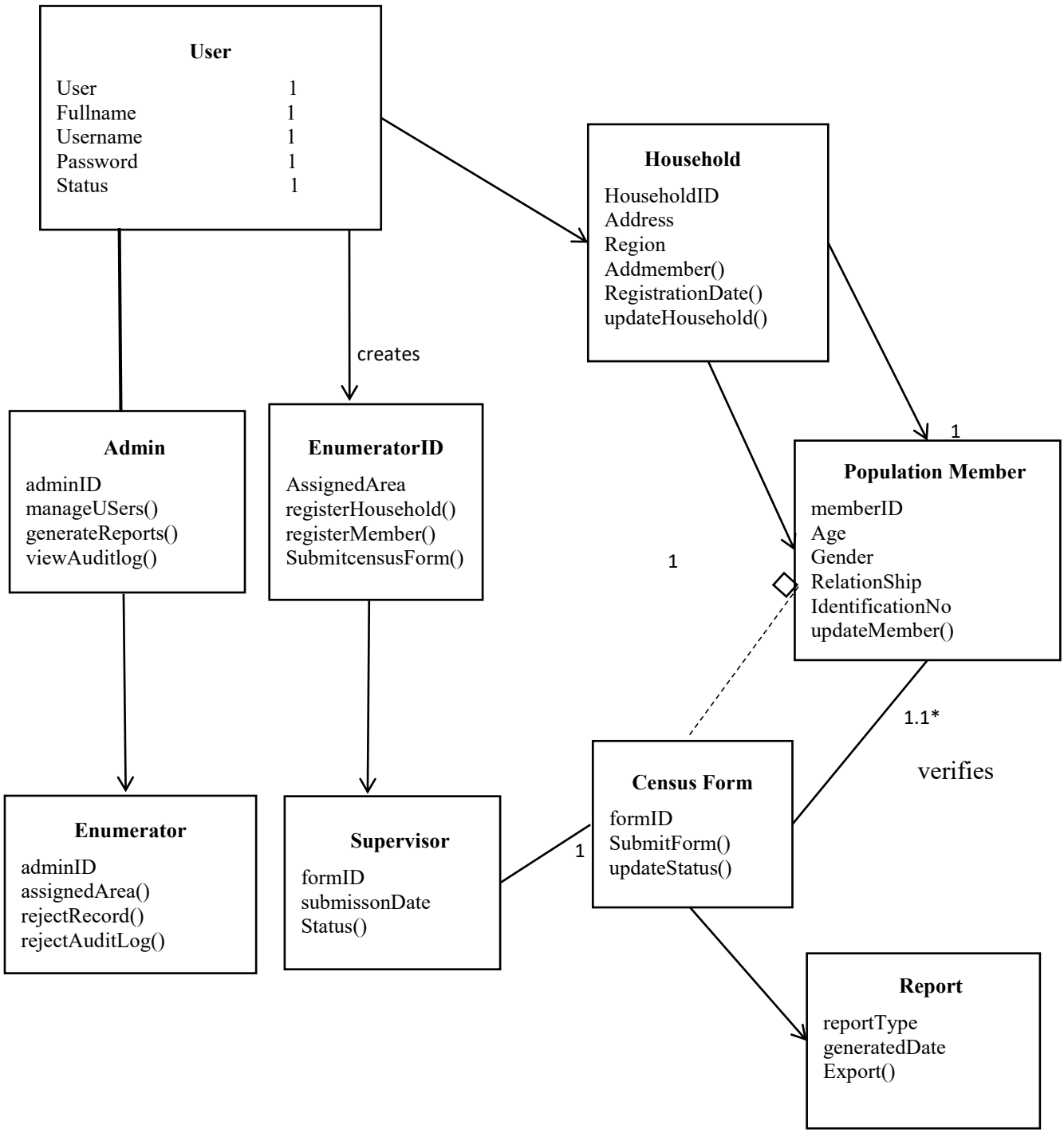
Operations:

- generate()
- export()

Class Relationships

- User is a parent class of Admin, Enumerator, Supervisor, and Citizen (inheritance).
- The enumerator is the person who creates records for households and also for the people who live in those households, which are called population members. The enumerator creates these records for households and, for each population member.
- A household is connected to population members. This means that one household can have lots of population members living in it. The household has a connection, to each of these population members.
- The Census Form is related to one household and lots of population members. The Census Form deals with one household and many population members.
- Supervisor verifies CensusForm.
- Admin generates Report.

Population Count system class diagram



4.6 Data Flow Diagram

The Data Flow Diagram represents the **flow of data** within the Population Count System. It shows how data is received from external entities, processed by the system, stored in data repositories, and returned as output. DFDs help in understanding how information moves through the system without focusing on technical implementation details.

Context Diagram

The context diagram provides a **high-level view** of the system as a single process interacting with external entities.

External Entities

- ✓ Admin
- ✓ Enumerator
- ✓ Supervisor
- ✓ Citizen

Process

- ✓ Population Count System

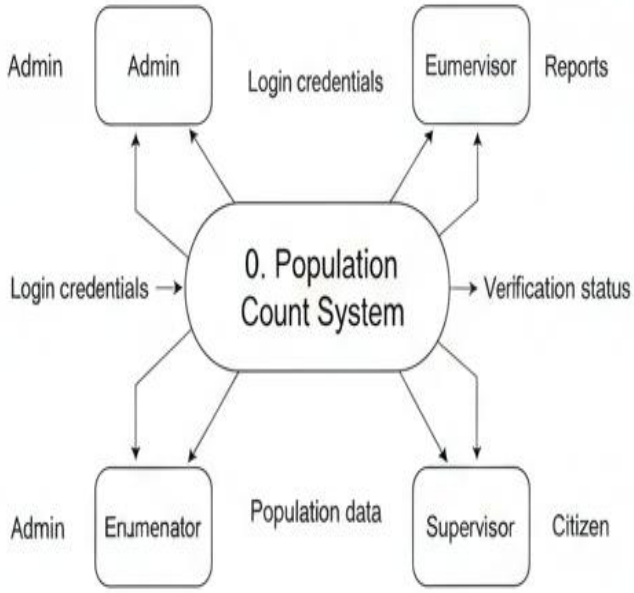
Data Flows

- ✓ Login credentials
- ✓ Population data
- ✓ Verification status
- ✓ Reports

Context Diagram Description

All external entities interact with the Population Count System by sending and receiving data. The system processes inputs such as census information and user requests, and returns outputs such as confirmation messages and reports

A high-livenl view of the system's interaction with external entiies



Level 1 DFD

The Level 1 DFD decomposes the system into major processes and data stores

Processes

- User Authentication
- Population Data Management
- Census Verification
- Report Generation
- User Management

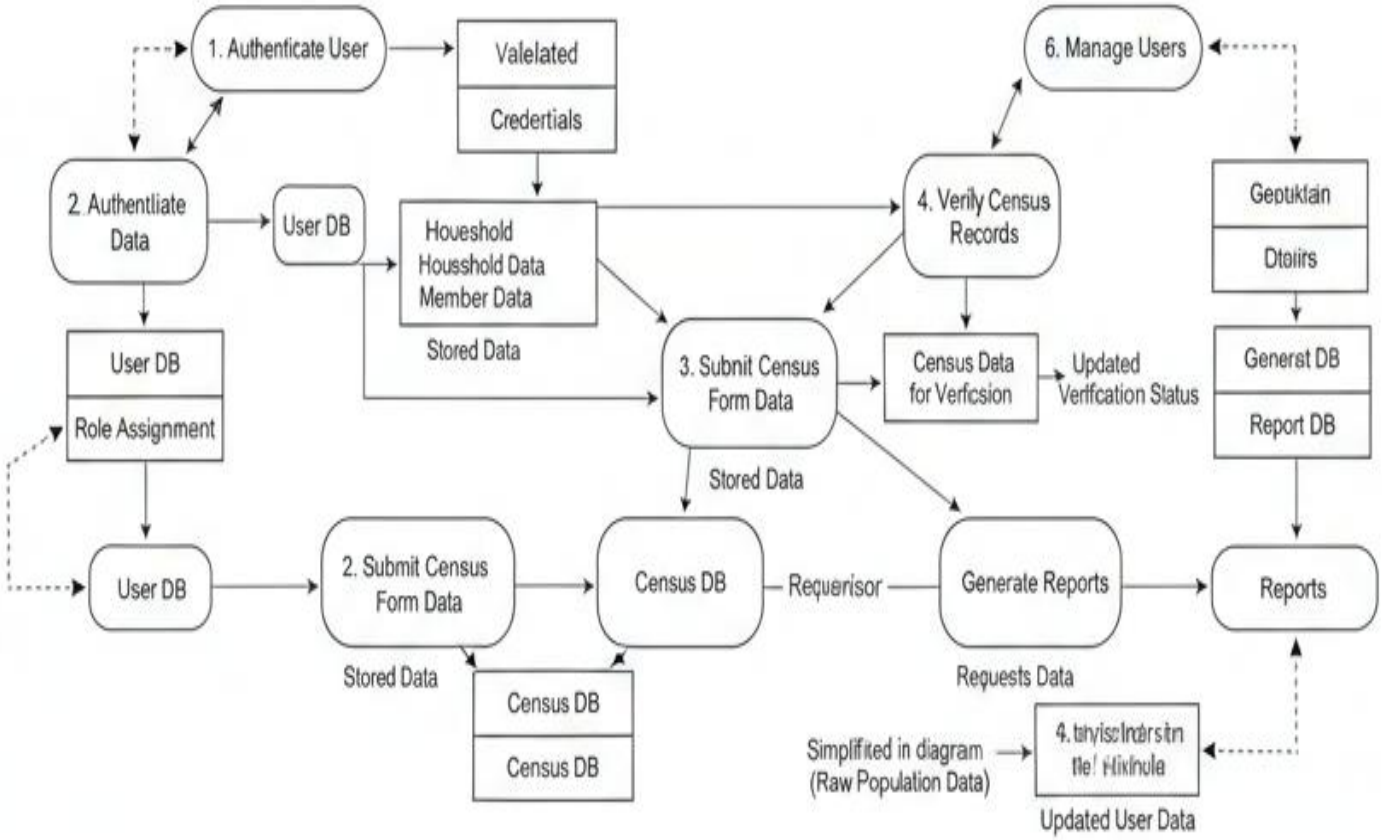
Data Stores

- User Database
- Household Database
- Population Database
- Census Form Database
- Report Database

Level 1 DFD Description

- ✓ Users submit login credentials to the authentication process.
- ✓ Enumerators send household and population data to the population data management process.
- ✓ Census forms are stored and forwarded to the verification process.
- ✓ Supervisors update verification status.
- ✓ Admin generates reports using stored population data.
- ✓ All processed data is stored in the respective databases.

Data Flow Diagram : Population Count System
Level 1: -



DFD Components Summary

Processes

- ✓ Authenticate User
- ✓ Manage Population Data
- ✓ Submit Census Form
- ✓ Verify Census Records
- ✓ Generate Reports
- ✓ Manage Users

Data Stores

- ✓ User DB
- ✓ Household DB
- ✓ Population DB
- ✓ Census DB
- ✓ Report DB

Data Flows

- ✓ Credentials
- ✓ Household Data
- ✓ Member Data
- ✓ Verification Status
- ✓ Reports

4.7 Entity Relationship Diagram(ERD)

The Entity Relationship Diagram (ERD) describes the logical structure of the database for the Population Count System. It identifies the main entities, their attributes, and the relationships between them. The ERD ensures data is stored efficiently, avoids redundancy, and maintains data integrity

Entities and Attributes

User

- ✓ user_id (PK)
- ✓ full_name
- ✓ username
- ✓ password
- ✓ role
- ✓ status

Enumerator

- ✓ enumerator_id (PK)
- ✓ assigned_area
- ✓ user_id (FK)

Supervisor

- ✓ supervisor_id (PK)
- ✓ user_id (FK)

Household

- ✓ household_id (PK)
- ✓ address
- ✓ region
- ✓ registration_date
- ✓ enumerator_id (FK)

PopulationMember

- ✓ member_id (PK)
- ✓ full_name
- ✓ age
- ✓ gender
- ✓ relationship
- ✓ national_id
- ✓ household_id (FK)

CensusForm

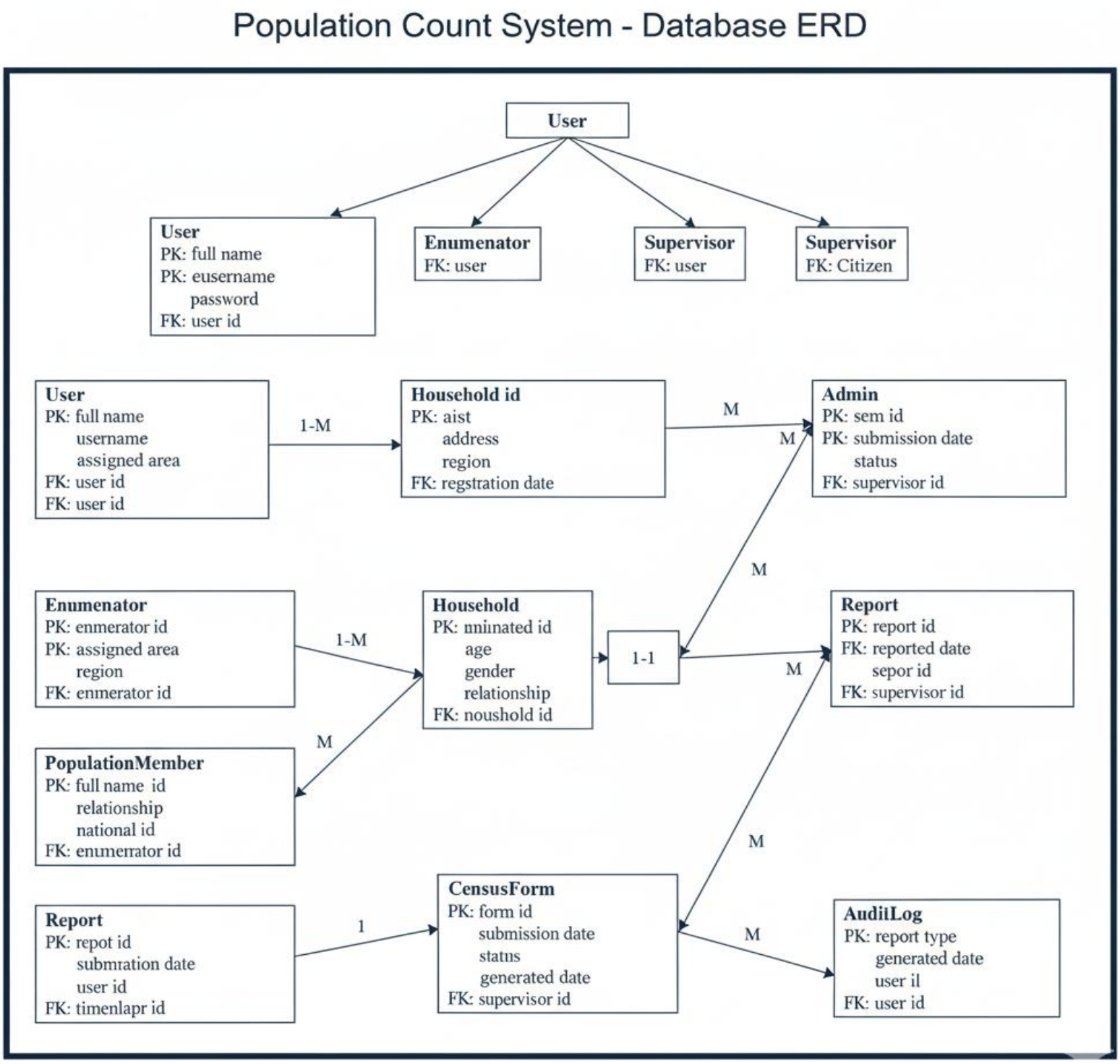
- ✓ form_id (PK)
- ✓ submission_date
- ✓ status
- ✓ household_id (FK)
- ✓ enumerator_id (FK)
- ✓ supervisor_id (FK)

Report

- ✓ report_id (PK)
- ✓ report_type
- ✓ generated_date
- ✓ user_id (FK)

AuditLog

- ✓ log_id (PK)
- ✓ action
- ✓ timestamp
- ✓ user_id (FK)



Entity Relationships

- A User can be an Enumerator, Supervisor, Admin, or Citizen.
- An Enumerator registers many Households.
- A Household contains many Population Members.
- A Household is linked to one Census Form.

- An Enumerator submits many Census Forms.
- A Supervisor verifies many Census Forms.
- An Admin generates many Reports.
- A User can have many Audit Log records

Relationship Summary Table

Entity A	Relationship	Entity B
User	1 — 1	Enumerator
User	1 — 1	Supervisor
Enumerator	1 — M	Household
Household	1 — M	PopulationMember
Household	1 — 1	CensusForm
Enumerator	1 — M	CensusForm
Supervisor	1 — M	CensusForm
User	1 — M	Report
User	1 — M	AuditLog

4.8 User Interface Requirements

The Population Count System is made to be easy to use. It is simple and straightforward for all the people who use it. The Population Count System helps people collect information, about the population check that it is correct manage the data and make reports. The Population Count System does all this while reducing mistakes that people might make and making it easier for them to use the Population Count System.

The system will be something that people can use on the internet. It will be like a website that people can get to with any web browser. They can use the system on the internet with a web browser, like the system is a website. The system is what people will use

General UI Requirements

The interface will show things to different people. It depends on what kind of user you're. For example if you are an Admin you will see features. If you are an Enumerator you will see features. The same thing goes for a Supervisor or a Citizen. The interface is set up to show features based on the type of user whether that is an Admin, an Enumerator, a Supervisor or a Citizen.

The system needs to have navigation menus that're easy to understand and a layout that looks the same on every single page of the system. This way the system will be simple to use. The navigation menus of the system should be clear. The layout of the system should be consistent, across all pages of the system.

When we fill out forms to enter data these forms should check what we are typing. Make sure it is correct. If we make a mistake the data entry forms should show us an error message so we know what we did wrong. This is really important, for all data entry forms.

The interface will give you a message when you do something. This message is to let you know that what you did was successful. The interface is supposed to do this for all actions like when you complete a task or make a change. This way you can be sure that the interface is working properly and that your actions were successful.

The system needs to work on desktop computers and tablet devices so it has to be able to change its layout to fit different screen sizes. This means the system has to support design, for desktop and tablet devices.

The interface should be easy to understand. It should use simple language and fonts that are easy to read. The interface will have language and the fonts will be readable.

User-Specific Interface Requirements

- Admin Interface
- Dashboard displaying population statistics and summaries.
- User management screens for creating, updating, and deactivating users.
- Report generation interface with filtering options.

The interface that we use to look at the audit log is really important. We need to be able to see what is going on with the audit log. The audit log viewing interface has to be easy to use.

We have to make sure the audit log viewing interface works well so we can check the audit log whenever we want. The audit log is very important. The audit log viewing interface is a big part of that.

Enumerator Interface

- Login interface for secure access.
- Household registration forms.
- Population member data entry forms.
- Census form submission interface.

I want to see what is happening with the census records I sent in. The status view for submitted census records is what I am looking for. I need to check on the census records that I have already submitted.

Supervisor Interface

- Dashboard showing pending census records.
- We need to check the information that people give us for the census. So we have a page where we can say yes or no to this information. This page is like a gatekeeper for the census data. It helps us make sure the information is correct before we use it. We call this page the verification interface, for the census data. The verification interface is very important because it lets us approve or reject the census data.

- Record review screens with detailed household and member information.

Citizen Interface

- Login interface.
- View personal population data.
- Request data correction interface (if enabled).

Form Design Requirements

- We need to make sure that the mandatory fields are clearly marked so that everyone knows what they have to fill in. The mandatory fields will be marked in a way that you can easily see them.
- We will use drop-down menus for things, like gender and region and relationship because these're predefined values.
- We need to use date pickers when we have a field that's for a date. This way people can easily pick the date they want for the date fields. Date pickers are really helpful, for date fields because they make it simple to choose a date.
- The system will not let people send in a form if they do not fill out all the required fields. The system has to make sure that people fill out all the required fields before they can send in the form. If the required fields are missing the system will prevent the form from being submitted.
- When something is repeated the system will show a warning to let you know that there is a duplicate. The duplicate warnings will be displayed when they are needed for the user to be aware of the duplication. Duplicate warnings are important to notify the user of information.

Navigation and Feedback Requirements

- Navigation menus shall be persistent across pages.
- The breadcrumbs or the page titles should show you where you are, on the website. They help you understand the location of the webpage you are looking at which is the current location.
- Error messages shall be clear and user-friendly.
- Success messages shall confirm completed actions.

4.9 Logical Database Requirements

The logical database requirements define how data is organized, stored, and managed within the Population Count System. The database is designed to ensure data integrity, consistency, and efficient retrieval while supporting all system operations such as population registration, verification, reporting, and auditing.

The system shall use a relational database management system (RDBMS).

Database Design Requirements

- The database shall be designed based on the Entity Relationship Diagram (ERD).
- All tables shall have primary keys to uniquely identify records.
- Foreign keys shall be used to define relationships between tables.
- The database shall enforce referential integrity.
- Data redundancy shall be minimized through normalization (up to Third Normal Form – 3NF)
- The database shall support concurrent access by multiple users.

Logical Tables Overview

- ✓ User Table
- ✓ Stores system user accounts and access roles.
- ✓ user_id (PK)
- ✓ full_name
- ✓ username
- ✓ password
- ✓ role
- ✓ status

Enumerator Table

- ✓ Stores enumerator-specific information.
- ✓ enumerator_id (PK)
- ✓ assigned_area
- ✓ user_id (FK)

Supervisor Table

- ✓ Stores supervisor information.
- ✓ supervisor_id (PK)
- ✓ user_id (FK)

Household Table

Stores household registration data.

- ✓ household_id (PK)
- ✓ address
- ✓ region
- ✓ registration_date
- ✓ enumerator_id (FK)

Population Member Table

Stores individual population details.

- ✓ member_id (PK)
- ✓ full_name
- ✓ age
- ✓ gender
- ✓ relationship
- ✓ national_id
- ✓ household_id (FK)

Census Form Table

Stores census submission records.

- ✓ form_id (PK)
- ✓ submission_date
- ✓ status
- ✓ household_id (FK)
- ✓ enumerator_id (FK)
- ✓ supervisor_id (FK)

Report Table

Stores metadata for generated reports.

- ✓ report_id (PK)
- ✓ report_type
- ✓ generated_date
- ✓ user_id (FK)

Audit Log Table

Tracks system activities for security and accountability.

- ✓ log_id (PK)
- ✓ action
- ✓ timestamp
- ✓ user_id (FK)

Constraints and Rules

- National ID shall be unique for each population member.
- Deletion of users shall be restricted if linked records exist.
- Census forms shall not be modified after verification.
- Only authorized roles may update sensitive data.
- Indexing and Performance Requirements

Indexes shall be created on frequently searched fields such as:

- ✓ national_id
- ✓ household_id
- ✓ user_id

Database queries shall be optimized for report generation.

Data Security Requirements

- Passwords shall be stored in encrypted or hashed format.
- Access to tables shall be controlled based on user roles.
- Audit logs shall be immutable.

4.10 Quality Attributes

The Population Count System has some qualities that are really important. These are the things that make the Population Count System work well and keep people happy with it. The Population Count System needs to be good at what it does and work properly when it is being used. The Population Count System has to be reliable and make people satisfied, with the Population Count System. This means the Population Count System can handle life situations and keep working for a long time.

<u>Performance</u>

The system needs to give people answers when they are using it like when they are putting in data or checking to see if everything is correct or when they are making reports.

The system also needs to be good, at getting population data and making reports so it can handle a lot of information without slowing down. Population data and report processing need to be optimized so the system can handle datasets efficiently for population data retrieval and report generation.

Reliability

The system needs to store population data in an accurate way. All the data that people submit has to be checked to make sure it is correct before it is stored. The system has to be able to handle mistakes so that population data is not lost or damaged. The population data is very important. The system must keep it safe. The system will make sure that the population data storage is done correctly.

Availability

The system will be available to authorized users when we are doing census operations and it will not be down, for a time. We will make sure to have backups of the database and ways to get the system working again if something goes wrong so the system keeps running.

Usability

The user interface should be easy to use and understand. It needs to be simple so that people who do not know a lot about technology can use it. The Facebook user interface is an example of this. Users with technical knowledge will be able to do their jobs well because the user interface has clear forms and feedback messages. The user interface also has navigation which helps the user interface. The user interface is role-based which means the user interface is designed for jobs. The user interface is simple and easy to use which is good, for users who use the user interface.

Security

The system has to make sure that only the right people can do things this is called role-based access control. We need to keep user credentials safe and sound. We have to protect sensitive data from people who should not see it. The system will write down all actions in something called audit logs.

This way we can keep track of what's happening with the system and the sensitive data. The system will use role-based access control to keep everything.

Scalability

The system needs to work for more and more users, households and population records. It should not get slower when it has to handle all these users and population records.

The system is made in a way that it can be expanded to cover areas. This means the system can support the population records and households in geographic regions. The system will keep working for the users, in these larger geographic regions.

Maintainability

The system will be simple to update and modify because it has an architecture. This means that if we make changes to one part of the system it will not hurt the parts of the system. We can make changes to one component. It will not mess up the other components of the system.

Data Integrity

The system needs to keep the data accurate and consistent. It does this by using validation rules and database constraints. It also has controlled update mechanisms. This way the system can. Deal with duplicate and invalid records. The system will handle these records in a way. The system will make sure that the data is correct and consistent by using these methods to keep the data accurate and handle any problems, with invalid records.

Interoperability

The system shall allow data export in common formats such as PDF and Excel to support external analysis and reporting by government agencies.

4.11 Requirements Traceability Matrix (RTM)

The Requirements Traceability Matrix (RTM) establishes a clear relationship between **system requirements and design artifacts**. It ensures that every requirement is properly addressed, implemented, and verifiable throughout the system development lifecycle.

Requirements Traceability Matrix

Requirement ID	Requirement Description	Related Use Case	Design Artifact	Verification Method
FR-01	User authentication	Login	Sequence Diagram, Class Diagram	Functional Testing
FR-02	Register household	Register Household	Activity Diagram, ERD	Functional Testing
FR-03	Register population member	Register Population Member	Class Diagram, ERD	Functional Testing
FR-04	Submit census form	Submit Census Form	Activity Diagram, Sequence Diagram	Functional Testing
FR-05	Verify population records	Verify Population Records	Activity Diagram, Sequence Diagram	Inspection
FR-06	Update population data	Update Population Data	Class Diagram, DFD	Functional Testing
FR-07	Detect duplicate records	Duplicate Detection	DFD, Logical DB Design	Data Validation Testing
FR-08	Generate population reports	Generate Reports	Sequence Diagram, DFD	Functional Testing
FR-09	Manage users and roles	Manage Users	Class Diagram, UI Design	Inspection
FR-10	Maintain audit trail	View Audit Trail	ERD, Logical DB Design	Inspection
NFR-01	System performance	All Use Cases	Architecture Design	Performance Testing
NFR-02	Data security	All Use Cases	Security Design, DB Design	Security Testing
NFR-03	Usability	All Use Cases	UI Requirements	Usability Testing

NFR-04	System scalability	All Use Cases	System Architecture	Load Testing
NFR-05	Data integrity	All Use Cases	ERD, DB Constraints	Data Integrity Testing

RTM Summary

The Return To Work Meeting confirms that:

The system design is what supports all of the requirements. This means that the system design includes all the parts to make sure that the functional requirements of the system are met. The functional requirements are supported by the system design artifacts.

The company makes sure that all the things that the system needs to do but are not actually part of what the system does are taken care of when they design the system and think about its quality. They consider the architecture of the system and the quality of the system to address all -functional requirements. This means that the system is designed with these -functional requirements, in mind so that the system works well and meets the needs of the users and all of these non-functional requirements are properly taken care of through the architectural and quality considerations of the system.

Each thing that the system needs to do is something that we can check and make sure it works which means we are less likely to forget something that the system needs to do. The requirements are things that we can test and the requirements are things that we can verify so we can be sure that the requirements are met and the system does what it is supposed to do. This helps to reduce the risk of the system missing some functionality that the requirements need. The requirements are very important. We need to make sure that the system meets all of the requirements.

This traceability ensures the Population Count System is complete, consistent, and well-structured.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATIONS

This part of the book is about the thoughts on the Population Count System project. It tells us what the Population Count System project did well and what can be done to make the Population Count System better, in the future. The Population Count System project had some results and there are some things that can be added to the Population Count System to make it even better.

5.1 Conclusion

The Population Count System was made to help people collect and manage information about the population. It makes sure the data is correct and easy to use. This system is better, than counting people by hand because it uses computers to keep track of the information. The Population Count System helps people get the numbers and it is easy to access the information. It is a way to collect and report population data. The Population Count System is a solution because it is organized and it helps people do their jobs more efficiently.

The project had a lot of requirements that needed to be met. We had to make sure the system worked the way it was supposed to and that it was also good at doing what it was not supposed to do. To make this happen we created a plan for the system. We used a lot of diagrams to map out how the system would work.

We made use case diagrams, activity diagrams, sequence diagrams, class diagrams, data flow diagrams and entity relationship diagrams. These diagrams really helped us understand how the system would behave and how the data would be organized.

The models we created made sure that what the system was supposed to do and how it was actually built were the same. The system requirements and the implementation design were consistent which is very important for a project, like this.

The system helps people who collect information about households and people do their job quickly. It also helps the people in charge check the records and the people who manage the system take care of the users and make reports with statistics.

The way the database is set up makes sure the information is correct and not repeated and the part that users interact with is simple and easy to use for everyone depending on what their job's, like the household and population data collectors, supervisors and administrators.

The Population Count System is really good at doing what it is supposed to do. It gives us a safe and easy to use platform. This platform helps us manage population data and make decisions. The Population Count System is also very good, at handling a lot of information and growing with our needs.

The Population Count System supports population data management and decision-making for the Population Count System. This is very important for the Population Count System to work properly.

5.2 Recommendations

The Population Count System does what it is supposed to do. However there are some things that can be added to the Population Count System to make it work better. This will help the Population Count System in the future. The Population Count System needs these changes to improve the Population Count System.

The system can be made bigger to include support for applications. This means that people who collect data can do their job offline with applications and then sync the data when they have internet. This will make the system more useful in places that're really far away, from cities. The mobile applications will really help the system work better in these areas.

We can use computer programs to look at population trends. These programs can show us things like how the population is growing, where people are moving to and, from and what the population looks like in terms of age and other things.

Population trends are very important. These tools can help us understand them better. We can get an idea of population trends by using these computer programs to look at the data. Population trends are what we are trying to understand.

The system can be made better by adding a way to check peoples identities using information or their national ID. This will help reduce duplicate records and make sure the data is correct. The system will be more accurate because it will use ID verification and biometric verification to check peoples identities. This means the system and national ID verification and biometric verification will work together to make data more accurate.

We need to make sure our system is really secure. So we can add some security features. Things like two-factor authentication and encrypted data transmission can help. This will make our system security stronger. We should do this to protect our system. Two-factor authentication and encrypted data transmission are important, for system security.

Finally, the system can be integrated with other government information systems to support broader administrative planning and policy development.

REFERENCES

✧ **Ethiopia Population & Housing Census – CSA / ESS (official)**

This is the main *government* source for population data in Ethiopia.

Ethiopian census official portal:

https://ess.gov.et/census/

This page lists census files and official census reports (Population & Housing Census) published by the Ethiopian Statistical Service .

✧ **United Nations Principles & Recommendations for Population and Housing Censuses**

This is the official international standard document guiding census practices worldwide.

UN Principles & Recommendations (Revision 3 PDF):

https://unstats.un.org/unsd/demographic-social/Standards-and-Methods/files/Principles_and_Recommendations/Population-and-Housing-Censuses/Series_M67rev3-E.pdf

This is the official UN document with international census standards you can cite.

✧ **Ethiopian National Population Policy (official policy document)**

This is a *real* Ethiopian government policy document establishing population policy context.

National Population Policy of Ethiopia (English PDF):

https://eidm.org/sites/default/files/2022-

06/Population%20Policy%20%28English%29.pdf

This is an official-looking English version of the policy.

- The tabular format used in this project follows the templates provided during the course lectures

- Some small structuring tips and sanity checks were suggested by a virtual assistant. Any remaining confusion is purely our own responsibility